

PRODUCT NETWORKS AND THE COMPETITIVE EFFECTS OF ACQUISITIONS*

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Abstract

We develop a network-based screening framework to evaluate the effects of horizontal acquisitions. Leveraging tools from network theory, we show that the effect of a change in ownership can be locally decomposed into an upward-pricing-pressure component and a network-centrality component that captures how pricing pressure propagates through the system of competing products. For full mergers, the change in network-centrality summarizes the unilateral effect, yielding a screening statistic that can be computed with data commonly available in merger investigations. Using price and quantity data from the European automobile industry, we show that the network model matches substitution patterns and simulated merger effects from a random-coefficients nested-logit benchmark. We further show that changes in centrality provide substantial incremental explanatory power for predicted price effects beyond both the Gross Upward Pricing Pressure Index and changes in concentration. In screening exercises, we quantify the gains from complementing traditional screening tools with centrality-based rules.

Keywords: Diversion Ratio, Network Centrality, Antitrust, Merger Screening.

JEL Classification: D43, G34, L13, L41

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1 Introduction

The analysis of horizontal mergers requires tools that are both tractable and economically grounded. Traditional concentration indices, such as the Herfindahl-Hirschman Index (HHI) and its change (ΔHHI), are easy to compute and widely used in practice. Yet they do not capture the closeness of competition when products are differentiated and require an ex ante market definition, a step that is often difficult and frequently contested.¹ Structural demand models in the tradition of [Berry et al. \(1995\)](#) offer a richer, micro-founded way to recover substitution patterns and simulate merger effects. However, they require substantial data and are computationally intensive, which limits their scalability and practical use. Moreover, substitution patterns remain tied to modeling choices: estimates depend on the specified choice set and, possibly, on the chosen nesting structure, whose misspecification can lead to biased substitution patterns and merger predictions ([Fosgerau et al., 2024](#)).

Between concentration screens and full merger simulations lies a class of first-order approaches (FOAs), such as the upward pricing pressure (UPP) and the Gross Upward Pricing Pressure Index (GUPPI). These approaches summarize the direct pricing incentive created by internalizing diversion between the merging products, without solving for the full post-merger equilibrium. They are attractive because they do not require an explicit market definition and can often be implemented using information on the merging products’ margins and diversion ratios. However, these inputs are not always easy to estimate accurately, and the resulting measures are local by design: they capture the direct incentive to raise prices created by common control, but not the strategic responses of rival products in the market.

This paper develops a screening framework for horizontal acquisitions that bridges FOAs and merger simulations while remaining tractable and scalable. We represent product-market competition as a network whose links are determined by diversion ratios.² As in traditional FOAs, the predicted price effects capture the direct upward pricing pressure generated when merging firms internalize diversion between their products. Unlike standard FOAs, however, the framework also accounts for how this initial pricing pressure propagates through the broader system of differentiated products, as in merger simulations.

We first consider a minority stake acquisition and show that the price effect decomposes into a UPP-style component and a network-centrality term (Proposition 1). The first captures the direct diverted-margin incentive created by the acquisition.

¹As stated by Prof. Jonathan Baker, “the outcome of more cases has surely turned on market definition than on any other substantive issue.” See <https://project-disco.org/competition/090518-antitrust-in-60-seconds-market-definition/>. See also [Kaplow \(2015\)](#).

²The diversion ratio from product j to product k is the fraction of consumers who leave product j after a price increase and switch to product k .

The second summarizes how this incentive is transmitted through the diversion network following competitors’ pricing responses. We then turn to full mergers and derive closed-form expressions for price effects, consumer surplus changes, and CMCRs (Proposition 3). The impact of a merger is fully characterized by the induced change in the competition network and thus the centrality of each product.

These results imply that our framework also delivers a centrality-based screening statistic. Unlike the HHI and ΔHHI , whose theoretical justification for merger screening relies on aggregative environments where substitution patterns are primarily driven by market shares (Nocke and Whinston, 2022), our centrality approach naturally admits a closeness measure of competition between products and hence allows for flexible substitution patterns. This flexibility is of great importance because substitution patterns can vary substantially across products within the same industry.³

One additional advantage of our approach is that it naturally accommodates arbitrary changes in the ownership structure, including those implied by incumbents’ acquisitions of minority stakes in rival firms or institutional investors’ common ownership. Recent research shows that even small shareholdings may soften competition (Azar et al., 2018; López and Vives, 2019), and enforcement agencies have begun scrutinizing such deals more closely, as reflected in the 2023 U.S. Merger Guidelines.⁴ Existing tools, however, offer limited guidance: modified concentration indices still rely on an ex ante market definition, and structural demand models become difficult to interpret and implement when ownership links proliferate.

To bring our framework to the data and characterize the diversion network more explicitly, we draw on the literature on hedonic demand models (Lancaster, 1966; Rosen, 1974; Epple, 1987; Feenstra and Levinsohn, 1995) and build on the linear-quadratic hedonic specification in Pellegrino (2025) which allows us to map the diversion-ratio-based product network into a characteristics-based network.

Within this linear-quadratic specification, the cross-price elasticity between two products, and hence their diversion ratio, is proportional to a weighted inner product of their attribute vectors (Proposition 5). Consequently, acquisition-induced changes in centrality can be approximated directly from observable product characteristics. This implementation builds on the insight that products’ relative positions in characteristic space are informative about substitution (Gandhi and Houde, 2023), and provides a parsimonious way to estimate heterogeneous substitution patterns. In many industries, information on products’ characteristics is available to agencies or can be collected at

³For example, in the beer market, Fan and Yang (2025) documents limited substitution between craft and non-craft beers despite their presence in the same product category. As a result, ownership changes across these segments may generate small price effects even when concentration measures change substantially.

⁴Guideline #11 states: “When an Acquisition Involves Partial Ownership or Minority Interests, the Agencies Examine Its Impact on Competition.”

scale from public sources, such as websites and other unstructured sources. Recent advances in large language models (LLMs) have sharply reduced the cost of collecting and organizing these data, making it feasible to implement centrality-based merger screening without requiring agencies to obtain additional information from firms or other market participants.

Lastly, we apply the framework to the European automobile industry and estimate the network model using price–quantity data. We show that the demand parameters are identified and can be estimated using a simple linear IV strategy, under the assumption that unobserved characteristics shift utility additively. The resulting estimates imply a diversion matrix determined by the network representation of product characteristics, which in turn maps directly into the centrality-based markup system.

Comparing the demand estimates obtained from this framework with those we obtain from traditional discrete-choice models, we find that the network model captures economically meaningful substitution patterns in the data. In particular, it generates diversion ratios similar to those obtained from a Random Coefficients Nested Logit (RCNL) model, capturing stronger substitution within car segments without imposing a nesting structure *ex ante*. When we simulate all possible mergers across European car manufacturers, we find that predicted price and consumer surplus effects are highly correlated and relatively similar when we rely on estimates from the network model and the RCNL, suggesting that our approach captures the key substitution forces driving unilateral effects.

We also evaluate the role of centrality as a screening statistic. In the automobile industry, changes in centrality provide substantial incremental explanatory power for predicted price effects beyond both GUPPI and changes in HHI. This indicates that our index captures information about merger harm that is not summarized by concentration measures or by traditional first-order screens based on direct diverted margins alone. In particular, when price effects are simulated using the RCNL benchmark, adding ΔHHI to GUPPI raises the R^2 from 0.765 to 0.814, whereas adding $\Delta \text{Centrality}$ raises it to 0.855, which is nearly the same explanatory power as the specification that includes all three screening statistics. This evidence directly supports the idea that the network-propagation component emphasized by our framework is economically significant in the data.

We then assess the performance of centrality-based screening in a policy decision framework in which the regulator seeks to maximize expected consumer surplus from approved mergers. Using demand estimates from the RCNL model, we simulate all possible mergers in the sample under alternative efficiency assumptions (2%, 5%, and 10%) and compare decision rules based on changes in centrality with concentration-based presumptions and GUPPI. We find that centrality-based rules improve on the

2023 concentration-based presumption across enforcement regimes, while their advantage relative to other screening tools depends on the underlying efficiency environment.

When assumed efficiencies are small, the main screening challenge is avoiding the approval of harmful mergers. In the strict regime, which assumes a 2% marginal-cost efficiency, screening on $\Delta\text{Centrality}$ reduces the share of welfare-reducing mergers that are approved by 24.9 percentage points relative to the concentration-based presumption and by 57.7 percentage points relative to GUPPI. A simple $\Delta\text{HHI} > 100$ rule also performs well in this regime, even better than the $\Delta\text{Centrality}$ statistic, but it becomes overly restrictive once larger efficiencies are allowed.

In the baseline regime, which assumes a 5% marginal-cost efficiency, screening on $\Delta\text{Centrality}$ raises consumer surplus from allowed mergers by 5.3% relative to the concentration-based presumption and by 2.8% relative to GUPPI, with even larger gains relative to the $\Delta\text{HHI} > 100$ rule.

In the lax regime, where efficiencies are assumed to be large, more permissive screens become relatively more attractive, and GUPPI performs slightly better than centrality-based screening. We also develop an estimated version of the $\Delta\text{Centrality}$ statistic that can be constructed using product characteristics alone. Although this measure is naturally less informative than its model-implied counterpart, it still performs well: for example, in the baseline regime, it increases consumer surplus from allowed mergers by 2.2% relative to the concentration-based presumption. Overall, this exercise quantifies the benefits of incorporating centrality-based rules as a complement to traditional merger screens.

Related Literature. A large literature has developed tools for evaluating the unilateral effects of horizontal mergers. Traditional concentration-based measures such as the HHI and ΔHHI remain widely used because they rely only on pre-merger market shares and are straightforward to compute and interpret. [Nocke and Whinston \(2022\)](#) provide a rationale for this practice by showing that, in a homogeneous-good Cournot model, a larger ΔHHI implies a larger CMCR required to offset the merger’s upward pricing incentives.⁵ However, in differentiated-product oligopoly models, the CMCR generally depends not only on concentration changes, but also on the merging parties’ shares and demand-side primitives such as elasticities and margins which concentration indices alone cannot capture.⁶

Merger simulation approaches ([Nevo, 2000](#)) incorporate detailed demand and cost

⁵[Bhattacharya et al. \(2023\)](#) show that, in the context of mergers of US consumer packaged goods manufacturers, price changes both across and within-merger are correlated with ΔHHI .

⁶For aggregative demand systems such as logit and CES, [Nocke and Whinston \(2022\)](#) show that the CMCR depends on a function of the merger partners’ shares other than ΔHHI itself, and illustrate numerically that ΔHHI can still be informative about merger impacts. More generally, [Nocke and Schutz \(2025\)](#) provide a comprehensive analysis for price-setting models with logit and CES demand, deriving conditions under which a merger raises consumer surplus.

systems and are typically more predictive of actual price changes (Illanes et al., 2025). However, they are computationally demanding and sensitive to functional-form assumptions, such as the curvature of demand (Crooke et al., 1999). This has led to interest in local measures derived from first-order conditions, such as upward pricing pressure indices, which summarize merging firms’ incentives to raise prices (Farrell and Shapiro, 2009, 2010; Jaffe and Weyl, 2013).⁷ These indices rely on estimating diversion ratios. Although their usefulness is well understood and emphasized in the 2010 U.S. merger guidelines, their practical application remains limited due to the challenges in measuring them empirically (Conlon and Mortimer, 2021): the most natural approach is to estimate a rich demand system, which allows flexible substitution, and use it to compute the implied diversion ratios, but this is difficult to do at scale.⁸ Recognizing this, a burgeoning literature has developed data-driven proxies for diversion ratios using second-choice survey data (Conlon et al., 2023), consumer churn (Qiu et al., 2024), and customer-overlap measures (Einav et al., 2026).

We contribute to this literature by developing a tractable, network-based framework for the analysis of horizontal acquisitions. Like traditional FOAs, our framework captures the direct pricing pressure created by the merging parties’ internalization of the diversion between their products. Like merger simulations, however, it also accounts for how this initial upward pricing pressure propagates through the broader system of differentiated products. We show that these propagation effects are summarized by the Bonacich centrality of the network and that, for full mergers, the change in centrality provides an effective screening statistic. In our empirical implementation, the diversion network is constructed from products’ positions in characteristic space. This yields a version of the centrality-based screen that can be implemented using product characteristics alone, which are often available to agencies and are likely to become increasingly accessible as recent advances in machine learning allow them to be extracted from unstructured data and used in demand estimation (Magnolfi et al., 2022; Bajari et al., 2023; Compiani et al., 2025).

We also contribute to the empirical industrial organization (IO) literature on oligopolistic competition with differentiated products. Most empirical IO models build on discrete-choice demand with price competition à la Bertrand on the supply side (Berry, 1994; Berry et al., 1995).⁹ These approaches flexibly incorporate unobserved

⁷Miller et al. (2016) and Miller et al. (2017) examine how well first-order condition measures predict the price effects that are found in merger simulations, finding that they work well when measures of pass-through are precise.

⁸Conlon and Mortimer (2021) further demonstrate that there is no single measure of diversion, just as there is no single measure of elasticity. Estimates of both diversion and elasticity depend on where one evaluates the demand curve.

⁹Bresnahan (1987) was among the first to estimate a discrete choice model of oligopolistic competition with products that are differentiated along one dimension (i.e., à la Hotelling). For an exception with continuous demand, see Dubois et al. (2014).

product characteristics and enable counterfactual merger analysis, but realistic substitution patterns typically require random coefficients and non-convex estimation, and the resulting relationships between characteristics, substitution, and markups are only implicitly characterized. Moreover, practitioners must often specify a nesting structure *ex ante* to capture appropriate substitution patterns, which may not always be straightforward and, if misspecified, can bias estimates (Fosgerau et al., 2024).¹⁰

This paper is closer in spirit to Feenstra and Levinsohn (1995), who link substitution patterns to distances in characteristic space, but differs by providing an *analytical* characterization of markups rather than an implicit one. By framing differentiated-product competition as a network game, we show that equilibrium markups can be expressed through Bonacich centrality and that cross-price elasticities correspond to weighted inner products of product attributes. This structure accommodates both substitutability and complementarity,¹¹ allows for unobserved characteristics, delivers intuitive substitution patterns, and can be estimated with straightforward linear IV methods.¹²

Lastly, our work relates to a growing literature that applies network theory to study oligopolistic competition. Following Ballester et al. (2006), several papers model differentiated-product competition as a network game. Ushchev and Zenou (2018) analyze price competition when varieties are located on a network and show that equilibrium prices are proportional to a sign-alternating Bonacich centrality. Galeotti et al. (2024) exploit the network structure of Slutsky matrices to characterize optimal tax–subsidy policies. Pellegrino (2025) develops a general equilibrium model in which firms compete à la Cournot or à la Bertrand, and product attributes induce a similarity network, providing empirical estimates of efficiency losses due to market power.¹³ We complement this line of work by showing how network representations of competition can be used to characterize the competitive effects of acquisitions and to construct tractable merger screening tools.

¹⁰An emerging literature is studying new approaches to infer nests directly from consumer behavior. For example, Atalay et al. (2023) use co-purchasing patterns.

¹¹See Gentzkow (2007) for a discrete-choice model with complementarities; see also Thomassen et al. (2017) and Lee and Allenby (2009) for quadratic-preference models across product categories.

¹²Product proximity also matters for estimation. Gandhi and Houde (2023) show that optimal price instruments should reflect distances in product characteristics. In our framework, this intuition follows directly from the closed-form markup system based on product similarity.

¹³Ederer and Pellegrino (2025) extends the framework to incorporate common institutional ownership. Moreover, Antón et al. (2023) employ the same general equilibrium model to show that the economy-wide effects of mergers are substantial even when firms are assumed to internalize common ownership concerns before the mergers.

2 A Network Approach to Equilibrium Analysis

We consider a market with J products available. The marginal (and average) cost of production is constant, and is denoted by c_j . The demand for product $j \in \{1, \dots, J\}$ at price vector $\mathbf{p} = (p_1, \dots, p_J)'$ is $q_j(\mathbf{p})$. We assume that the demand function is generated by utility maximization of a representative consumer with indirect utility function $V(\mathbf{p}, y)$, where y represents the income of this consumer. Therefore, the Jacobian of the demand function $S(\mathbf{p}) := d\mathbf{q}(\mathbf{p})/d\mathbf{p}'$ is symmetric.

In our subsequent analysis, we highlight the role of a network built upon the diversion ratio matrix, defined here:

$$D(\mathbf{p}) := I_J - [\text{diag}(S(\mathbf{p}))]^{-1}S(\mathbf{p}),$$

in determining the equilibrium prices and merger impacts. To see that this is the diversion ratio matrix, notice that $D_{ij} = 0$ for $i = j$ and $D_{ij} = -\frac{\partial q_j / \partial p_i}{\partial q_i / \partial p_i}$ for $i \neq j$ (using symmetry of $S(\mathbf{p})$), which is exactly the ratio of how much more of product j would be purchased to how much less i would be consumed when p_i is raised.

2.1 Independent Ownership

We start with the familiar independent ownership case, where each product is independently owned by a profit-maximizing firm, to illustrate the network approach to equilibrium analysis.

We consider Bertrand competition among the firms, where each firm independently and simultaneously sets the price for the product it owns. It is straightforward to derive the first-order condition (FOC):

$$\mathbf{q}(\mathbf{p}) + \text{diag}(S(\mathbf{p}))(\mathbf{p} - \mathbf{c}) = 0.$$

We assume that the first-order condition is also sufficient, and a unique solution $\mathbf{p}^*$ exists.

Since our interests are in comparative statics and merger analysis, we take the same approach as [Galeotti et al. \(2024\)](#) and work with a locally linearized (or locally linear) demand function around the equilibrium $(\mathbf{p}^*, \mathbf{q}^*)$. We abuse notation slightly and let S denote the Jacobian of $\mathbf{q}(\mathbf{p})$ evaluated at $\mathbf{p}^*$. Similarly, let D denote the diversion ratio evaluated at $\mathbf{p}^*$. Let $\boldsymbol{\alpha} := -S^{-1}\mathbf{q}^* + \mathbf{p}^*$, then the linearized demand around the original equilibrium is $\mathbf{q} = -S(\boldsymbol{\alpha} - \mathbf{p})$.

With this linearization, we are ready to show that equilibrium markups admit an intuitive network-based decomposition.¹⁴ For that purpose, we introduce the Bonacich

¹⁴This decomposition is exact when consumers have linear-quadratic preferences and can also be

centrality measure, which is defined below following [Jackson \(2008\)](#).

Definition 1 Consider a network with J nodes, adjacency matrix A , node weights $\mathbf{u}$, and attenuation factor $\delta > 0$. The J -vector of (weighted) Bonacich centralities $\mathbf{b}(A, \delta, \mathbf{u})$ is given by

$$\mathbf{b}(A, \delta, \mathbf{u}) := (I_J - \delta A)^{-1} \delta A \mathbf{u} = \sum_{k=1}^{\infty} \delta^k A^k \mathbf{u}. \quad (1)$$

The j -th element of $\mathbf{b}(A, \delta, \mathbf{u})$ summarizes how central node j is in the network. This measure of centrality is widely used in social networks because it captures a node's importance in terms of how connected this node is to others and how connected the nodes it is connected to. According to the definition of Bonacich centrality, a node's importance is a weighted sum of the walks that emanate from it. Moreover, if $\delta \in (0, 1)$, walks of shorter length are weighted more.¹⁵

The following equilibrium characterization result ensues.

Lemma 1 *Around the original equilibrium, the new equilibrium markup after an ε -perturbation (on, e.g., $\mathbf{c}$ or $\boldsymbol{\alpha}$) is*

$$\mathbf{p}^\varepsilon - \mathbf{c}^\varepsilon = \frac{\boldsymbol{\alpha}^\varepsilon - \mathbf{c}^\varepsilon}{2} - \mathbf{b}\left(D, \frac{1}{2}, \frac{\boldsymbol{\alpha}^\varepsilon - \mathbf{c}^\varepsilon}{2}\right).$$

Notably, $\frac{\alpha_j^\varepsilon - c_j^\varepsilon}{2}$ is the markup charged by a monopolist facing linear inverse-demand $p_j = \alpha_j - S_{jj}^{-1} q_j$. This decomposition is closely related to the ones derived by [Chen et al. \(2018\)](#), [Ushchev and Zenou \(2018\)](#), and [Pellegrino \(2025\)](#). Our approach helps isolate the role of changes in the centrality measure in predicting the effects of mergers.

Lemma 1 highlights that the market power of each firm is shaped by its centrality in the competition network of all products: products that are more central face greater competitive pressure and therefore charge lower markups. Notably, the competition network takes the diversion ratio as the adjacency matrix, and the monopolist's markup (which is a measure of the standalone competition strength of each product) as the node weights. For product j , the competition pressure is higher if it is closer to (i.e., it has a high diversion ratio to) a strong competitor (with a high choke price or low cost), and therefore it must charge a lower markup.

The Bonacich centrality also captures multiple-step effects, which can be clearly seen from the walk expansion:

$$\mathbf{b}\left(D, \frac{1}{2}, \frac{\boldsymbol{\alpha}^\varepsilon - \mathbf{c}^\varepsilon}{2}\right) = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k D^k \left(\frac{\boldsymbol{\alpha}^\varepsilon - \mathbf{c}^\varepsilon}{2}\right).$$

derived in oligopoly models where firms compete on quantities ([Loseto, 2023](#)).

¹⁵This interpretation is motivated by the fact that when A is a binary $\{0, 1\}$ it k -th power A^k counts how many walks of length k are between any two nodes.

The term $D^k \left(\frac{\alpha^\varepsilon - c^\varepsilon}{2} \right)$ captures the influence of products that are k diversion steps away from a given product: a strong competitor disciplines its direct rivals' markups, which then disciplines their rivals, and so on. Hence, competitive pressure propagates along walks in the diversion network. The geometric decay factor $(1/2)^k$ reflects the linear-demand marginal-revenue tradeoff: a unit of competitive pressure transmitted one more step must pass through another best-response adjustment, and under linear demand each such adjustment is dampened by the same $1/2$ factor (as in the single-product monopoly rule $p - c = (\alpha - c)/2$). Therefore, more distant competitors matter less, with influence declining at rate $1/2$ per additional diversion step.

2.2 General Control or Ownership Structure

Our analysis can be easily extended to the case with a general ownership or control structure of the products. We introduce the decision weight matrix H to capture the extent to which the ownership structure makes the pricing decision for one product internalize the profit of another. Specifically, let

$$H_{jj} = 1, \quad H_{j\ell} \in [0, 1) \quad (\ell \neq j). \quad (2)$$

Product j internalizes a fraction $H_{j\ell}$ of product ℓ 's profit. That is, product j chooses p_j to maximize a weighted sum of profits:

$$\max_{p_j} \sum_{\ell=1}^J H_{j\ell} (p_\ell - c_\ell) q_\ell(\mathbf{p}). \quad (3)$$

This nests the standard independent ownership case when $H = I_J$. From the independent ownership, a merger between, say, firm 1 and firm 2, can be represented by $H = I + E_{12} + E_{21}$, where $E_{j\ell}$ is the matrix with all 0 entries except for an entry of 1 at row j , column ℓ .

This modeling approach allows flexibility in the analysis of common (partial) ownership and is also used in [Backus et al. \(2021\)](#) and [Ederer and Pellegrino \(2025\)](#). The decision matrix H can be micro-founded from common ownership as follows. Let β_{fs} denote shareholder (or fund) s 's cash-flow share in product j , and let γ_{js} be the corresponding governance weight that maps ownership into influence on price decisions. The ownership structure then implies a matrix of profit weights H with $H_{jj} = 1$ and

$$H_{j\ell} = \frac{\sum_s \gamma_{js} \beta_{\ell s}}{\sum_s \gamma_{js} \beta_{js}} \quad (j \neq \ell).^{16}$$

¹⁶Under the commonly used proportional-control assumption $\gamma_{js} = \beta_{js}$, these weights reduce to a normalized ownership overlap, $H_{j\ell} = (\beta'_j \beta_\ell) / (\beta'_j \beta_j)$.

which leads directly to the control matrix H , with $H_{j\ell} = \kappa_{j\ell}$. For our purposes, it suffices to work directly with the decision weight matrix H .

Given the decision weight matrix H , the FOC for the price competition game is

$$0 = \mathbf{q}(\mathbf{p}) + (H \odot S(\mathbf{p}))(\mathbf{p} - \mathbf{c}).$$

Again, we consider a local linearization of the equilibrium condition, with $\mathbf{q} = -S(\boldsymbol{\alpha} - \mathbf{p})$. We assume that $H \odot S$ is invertible and define the *internalized* diversion ratio matrix:

$$D_H := I_J - (H \odot S)^{-1}S = -(H \odot S)^{-1}[(E_J - H) \odot S], \quad (4)$$

where E_J is the $J \times J$ matrix with all 1s. Clearly, for the independent ownership case $H = I$, D_H is the standard diversion ratio matrix. At the other extreme when $H = E_J$ (i.e., every product fully internalizes the profit of all other products), we have $D_H = 0$. In general, $[(E_J - H) \odot S]$ is the non-internalized part of the demand diversion, and $-(H \odot S)^{-1}$ normalizes diversion with the internalized quantity effect (i.e., how much the quantity changes for internalized products), hence yielding a ratio. To further illustrate this definition, consider a two-product example. The diversion matrix, without considering profit internalization, is

$$D = \begin{pmatrix} 0 & D_{12} \\ D_{21} & 0 \end{pmatrix}.$$

Now suppose product 1 internalizes a fraction $h \in (0, 1)$ of product 2's profit (this can arise if the owner of product 1 acquires a *passive* financial stake h in product 2 without obtaining control: product 2 still chooses p_2 to maximize its own profit, while product 1 sets p_1 taking into account the profit it earns from its stake in product 2):

$$H = \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}.$$

Then the internalized diversion matrix in (4) becomes

$$D_H = \begin{pmatrix} h D_{12} D_{21} & (1 - h) D_{12} \\ D_{21} & 0 \end{pmatrix}.$$

Relative to D , the effective diversion link $1 \rightarrow 2$ is scaled down by $(1 - h)$, because the remaining fraction h is internalized. This internalization, in turn, generates a self-loop $1 \rightarrow 2 \rightarrow 1$ of size $h D_{12} D_{21}$ at node 1: When a higher p_1 shifts some demand to product 2, product 1 partially values the resulting gain at product 2 (that is, $h D_{12}$). Since this internalized gain is measured in product-2 units, D_{21} translates it into the

corresponding strength of feedback for product 1's own pricing incentive, yielding $h D_{12} D_{21}$.

The following result shows how the diversion network approach to equilibrium analysis extends to a general control structure.

Lemma 1' *Under a general ownership represented by the decision weight matrix H , the new equilibrium markup after an ε -perturbation (on, e.g., $\mathbf{c}$, $\boldsymbol{\alpha}$, or H) around the original equilibrium is*

$$\mathbf{p}^\varepsilon - \mathbf{c}^\varepsilon = \frac{\boldsymbol{\alpha}^\varepsilon - \mathbf{c}^\varepsilon}{2} - \mathbf{b}\left(D_{H^\varepsilon}, \frac{1}{2}, \frac{\boldsymbol{\alpha}^\varepsilon - \mathbf{c}^\varepsilon}{2}\right).$$

3 Network-Based Analysis of Acquisitions

In this section, we show how the diversion network approach provides new insight into the analysis of mergers and acquisitions. We start with the analysis of a small share acquisition and then move to the full merger case. For expositional ease, we first consider changes to the market structure from independent ownership, and then generalize the results to any control structure.

3.1 Minority Stake Acquisitions

In this part, we consider the impact of a local change to the control matrix $H = I_J$. Without loss of generality, we focus on a change to H_{12} , so that H becomes $I_J + \varepsilon E_{12}$. This can be understood as an acquisition of *passive* financial stake of product 2 by the owner of product 1. It is worth pointing out that, if a minority share acquisition is accompanied by proportional control, multiple elements of H would change at the same time. Nonetheless, our results easily extend to that case (or any other governance mapping), since the overall price or welfare impact—i.e., the total derivative with respect to multiple elements of H —is simply the sum of partial derivatives that we obtain below.

For later use, we introduce the Katz-Bonacich centrality for the J -node network with adjacency matrix A , node weights $\mathbf{u}$, and attenuation factor δ :

$$\boldsymbol{\kappa}(A, \delta, \mathbf{u}) \equiv (I_J - \delta A)^{-1} \mathbf{u} = \sum_{k=0}^{\infty} \delta^k A^k \mathbf{u}, \quad (5)$$

which is simply the Bonacich centrality plus the length-0 (i.e., self) term. Further, we define

$$\boldsymbol{\kappa}(j; D) := \boldsymbol{\kappa}\left(D, \frac{1}{2}, \mathbf{e}_j\right),$$

which is the discounted total length of all walks (of any length) from node j to each

node. This is a proximity measure of each node to j in the competition network: Node ℓ is “close” to node j if there are many or strong short paths from j to ℓ .

The following result gives the impact on prices and consumer surplus (CS) of the minority share acquisition.

Proposition 1 *When a minority share acquisition changes H from I_J to $I_J + \varepsilon E_{12}$, the price changes are:*

$$\left. \frac{\partial p_j}{\partial \varepsilon} \right|_{\varepsilon=0} = \frac{p_2 - c_2}{2} D_{12} \left[\mathbf{1}\{j = 1\} + b_j\left(D, \frac{1}{2}, \mathbf{e}_1\right) \right] = \frac{p_2 - c_2}{2} D_{12} \kappa_j(1; D). \quad (6)$$

By Roy’s identity, the CS impact is:

$$\left. \frac{\partial CS}{\partial \varepsilon} \right|_{\varepsilon=0} = -\frac{p_2 - c_2}{2} D_{12} \mathbf{q}' \boldsymbol{\kappa}(1; D). \quad (7)$$

The proposition reveals an intuitive network-based decomposition of the acquisition’s impact. For the price effects, the prefactor $\frac{p_2 - c_2}{2} D_{12}$ is the *direct upward pricing pressure*: firm 1 now partially captures the margin $p_2 - c_2$ on each unit diverted from product 1 to product 2 at rate D_{12} , with the factor $1/2$ reflecting the usual linear-demand best-response pass-through. In particular, this formalizes the intuition that the acquisition of a close substitute in terms of diversion ratio has a significant impact. The indicator $\mathbf{1}\{j = 1\}$ is the direct effect of a price increase for the acquirer. The Bonacich centrality term $b_j(D, \frac{1}{2}, \mathbf{e}_1)$ captures the competitive ripple effect that is the hallmark of the network approach: firm 1’s price increase propagates outward along walks in the diversion network—its direct rivals raise their prices, which in turn relaxes competition for *their* rivals, and so on—so that *every* product in the market is affected, with the impact on product j governed by its network proximity to product 1. By Roy’s identity, the impact on consumer surplus is the quantity-weighted sum of price impacts. Therefore, the total CS loss is larger when the acquirer is central to the largest sellers in the diversion network.

Next, we consider the cost synergy required to offset the CS loss.

Proposition 2 *When a minority share acquisition changes H from I_J to $I_J + \varepsilon E_{12}$, assuming that the acquirer and the target gain the same amount of cost reduction (i.e., c_j becomes $c_j - c$ for $j = 1, 2$), the cost synergy required for the minority share acquisition to be CS-neutral is:*

$$\frac{dc}{d\varepsilon} = (p_2 - c_2) D_{12} \frac{\mathbf{q}' \boldsymbol{\kappa}(1; D)}{\mathbf{q}' \boldsymbol{\kappa}(1; D) + \mathbf{q}' \boldsymbol{\kappa}(2; D)}. \quad (8)$$

The required synergy inherits a transparent network interpretation. The numerator $\mathbf{q}' \boldsymbol{\kappa}(1; D)$ is the CS harm from the acquisition (Proposition 1), capturing the *upward*

pricing pressure that originates at the acquirer and propagates through the diversion network. The denominator $\mathbf{q}'\boldsymbol{\kappa}(1; D) + \mathbf{q}'\boldsymbol{\kappa}(2; D)$ is the *downward pricing pressure* from a unit cost reduction shared equally by products 1 and 2, and it too propagates through the network via the same walk-based mechanism: a cost saving at product j lowers p_j , its rivals face intensified competition and reduce their prices, which then disciplines *their* rivals, and so on. The term $\mathbf{q}'\boldsymbol{\kappa}(j; D)$ measures the quantity-weighted reach of this pro-competitive cascade from node j , so the denominator reflects how broadly cost savings propagate based on the network proximity of each product to *both* the acquirer and the target. The ratio is therefore a number between zero and one that captures how much of the combined pass-through benefit is needed to neutralize the anticompetitive harm. Notably, this ratio depends *in opposite directions* on the network centrality of the acquirer and the target: if the target (product 2) is highly central—connected, via diversion, to many high-volume products—then $\mathbf{q}'\boldsymbol{\kappa}(2; D)$ is large, the denominator grows, and the synergy bar falls, because cost reductions at a central target propagate widely and generate large consumer benefits. Conversely, when the acquirer is central relative to the target, the ratio approaches one, and the required synergy is close to the full direct upward pricing pressure $(p_2 - c_2)D_{12}$.

3.2 Full Mergers

We now consider a full merger between firms 1 and 2. This means that the control matrix changes from I_J to $M_{12} := I_J + E_{12} + E_{21}$. We again consider the case of linear or linearized demand, assuming that the price impact is not so large that the linearization is invalid.

Proposition 3 *A merger between firms 1 and 2 changes prices by:*

$$\Delta \mathbf{p} = \mathbf{b}\left(D, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right) - \mathbf{b}\left(D_{M_{12}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right). \quad (9)$$

For the quantity-weighted price change $\mathbf{q}'\Delta \mathbf{p}$ (which approximates CS loss when price changes are small) to be 0, the required cost synergy (assumed to be the same for both firms) is

$$\Delta c = \frac{2\mathbf{q}'\left[\mathbf{b}\left(D, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right) - \mathbf{b}\left(D_{M_{12}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right)\right]}{\mathbf{q}'\boldsymbol{\kappa}(1; D_{M_{12}}) + \mathbf{q}'\boldsymbol{\kappa}(2; D_{M_{12}})}. \quad (10)$$

Just like in the minority share acquisition case, centralities in the diversion-based competition network play a key role in determining the merger impacts and the bar of compensating synergies. Equation (9) comes directly from the markup decomposition result (Lemma 1'), highlighting that a merger affects prices by altering the competition network and thus the centralities of each product. Equation (10)—its denominator in

particular—again reveals the insight that the downward pricing pressure from cost synergies propagates throughout the network from the merging parties.

3.3 Extensions and Discussions

The analysis of merger impacts through the lens of diversion network could enrich the antitrust authority’s toolbox and entails two major benefits. First, the centrality measures extend naturally to a general control structure. In contrast, the conventional screening tool, HHI index, is not easily defined when there is partial common ownership. Second, the computation of centrality measures does not require a full-fledged estimation of the demand system, and can be done with product characteristics data, which is relatively easy to obtain. We focus on the first point in this part, and develop the second point in detail in Section 4.

To show that the diversion network-based merger screening measures are applicable under a general initial market structure, we extend previous results accordingly. For a market structure change that perturbs the control matrix to $H^\varepsilon = H + \varepsilon\Delta H$, where ΔH is an arbitrary $J \times J$ matrix, it is useful to define

$$\omega_{\Delta H}(H) := -(H \odot S)^{-1}[(\Delta H \odot S)(\mathbf{p} - \mathbf{c})], \quad (11)$$

which measures the immediate price adjustment pressure (typically upward when $\Delta H \geq 0$) after the change in the internalization pattern H . The term $(\Delta H \odot S)(\mathbf{p} - \mathbf{c})$ aggregates, for each product j , the newly internalized “recapture” incentives created by the perturbation: a positive $\Delta H_{j\ell}$ makes the decision maker for p_j place more weight on product ℓ ’s margin $(p_\ell - c_\ell)$, and this incentive is stronger when the local substitution slope $S_{j\ell}$ indicates that changing p_j shifts demand toward ℓ . Just like in the definition of the internalized diversion ratio matrix D_H in (4), the prefactor $-(H \odot S)^{-1}$ *normalizes* these raw incentive shifts by the *internalized quantity effect* under the baseline structure H —that is, by how a price change for product j translates into quantity changes for the portfolio of products whose profits j internalizes. In this way, $\omega_{\Delta H}(H)$ expresses the direct incentive effect of ΔH in the same internalization-adjusted units that govern competitive interactions, so that it can be propagated through the diversion network in the subsequent equilibrium adjustment.

For minority share acquisitions that change H locally, we obtain the following.

Proposition 1’ *Consider a market structure change that perturbs the control matrix to $H^\varepsilon = H + \varepsilon\Delta H$, where ΔH is an arbitrary $J \times J$ matrix. Then the equilibrium price changes satisfy*

$$\left. \frac{\partial \mathbf{p}}{\partial \varepsilon} \right|_{\varepsilon=0} = \frac{1}{2} \kappa\left(D_H, \frac{1}{2}, \omega_{\Delta H}(H)\right). \quad (12)$$

By Roy's identity, the consumer-surplus effect is

$$\left. \frac{\partial CS}{\partial \varepsilon} \right|_{\varepsilon=0} = -\frac{1}{2} \mathbf{q}' \boldsymbol{\kappa} \left(D_H, \frac{1}{2}, \boldsymbol{\omega}_{\Delta H}(H) \right). \quad (13)$$

Proposition 2' Consider a market structure change that perturbs the control matrix to $H^\varepsilon = H + \varepsilon \Delta H$. The cost synergies required for the market structure change to be CS-neutral must satisfy:

$$\sum_{j=1}^J \mathbf{q}' \boldsymbol{\kappa}(j; D_H) \left(-\frac{dc_j}{d\varepsilon} \right) \Big|_{\varepsilon=0} = \mathbf{q}' \boldsymbol{\kappa} \left(D_H, \frac{1}{2}, \boldsymbol{\omega}_{\Delta H}(H) \right). \quad (14)$$

For more significant market structure changes, such as full mergers, assuming that the induced price changes are still within the range of linear approximation, we have the following.

Proposition 3' Consider an arbitrary change to the control matrix from H to $\widehat{H}$. The price changes (without any cost synergy) are:

$$\Delta \mathbf{p} = \mathbf{b} \left(D_H, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2} \right) - \mathbf{b} \left(D_{\widehat{H}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2} \right). \quad (15)$$

Suppose there are synergies that reduce costs from $\mathbf{c}$ to $\mathbf{c} - \Delta \mathbf{c}$ at the same time of the control matrix change. For the quantity-weighted price change $\mathbf{q}' \Delta \mathbf{p}$ (which approximates CS loss when price changes are small) to be 0, the required cost synergies must satisfy:

$$\frac{1}{2} \sum_{j=1}^J \mathbf{q}' \boldsymbol{\kappa}(j; D_{\widehat{H}}) \Delta c_j = \mathbf{q}' \left[\mathbf{b} \left(D_H, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2} \right) - \mathbf{b} \left(D_{\widehat{H}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2} \right) \right]. \quad (16)$$

4 Diversion Network and Product Attributes

Sections 2 and 3 established that equilibrium markups and the competitive effects of acquisitions are governed by the diversion ratio matrix D and its centrality structure. In those sections, we consider the local linearization of a general demand system $\mathbf{q}(\mathbf{p})$, and take its Jacobian S (and thus the diversion ratio matrix D) as given. The practical question for a regulator or analyst is: where does D come from, and what data are needed to compute it?

To make progress, we focus on the case of linear-quadratic preferences, where our local approximation is exact. The linear-quadratic framework has a long-standing tradition in differentiated oligopoly models (e.g., Singh and Vives (1984); Vives (1985)) because it implies a linear demand system with a diversion ratio that does not depend on quantities. To relate diversion ratios and product attributes, we further special-

ize to the hedonic-quadratic demand of [Pellegrino \(2025\)](#) where the quadratic utility term is defined as the outer product of the matrix of product attributes. Within this framework, the diversion ratio matrix can be obtained directly from observed product characteristics, without estimating a full demand system.¹⁷ Combined with the results of Sections 2 and 3, this means that the data requirements and computational burden of the centrality-based tools for merger screening are minimal.

4.1 Preferences and Demand

Products. There are J products available. Each product j is characterized by a vector of K attributes $\mathbf{x}_j = (x_{j1}, \dots, x_{jK})'$, collected in the $J \times K$ matrix X . The entry x_{jk} measures how much of attribute k a consumer obtains per unit of product j .

Utility. Following [Pellegrino \(2025\)](#), we consider a representative consumer that takes prices $\mathbf{p} = (p_1, \dots, p_J)'$ as given and chooses quantities $\mathbf{q} = (q_1, \dots, q_J)'$ and an outside good q_0 subject to $q_0 + \mathbf{p}'\mathbf{q} \leq y$. Preferences are

$$u(\mathbf{q}, X) = q_0 + \mathbf{q}'\boldsymbol{\alpha} - \frac{\beta}{2} \mathbf{q}'(I_J + \eta XX')\mathbf{q}, \quad (17)$$

where $\eta > 0$ governs the importance of product differentiation, $\boldsymbol{\alpha}$ (a J -vector) captures the consumer willingness to pay for each product, and $\beta > 0$ captures the consumer's preference for variety. The quadratic term $\mathbf{q}'(I_J + \eta XX')\mathbf{q}$ ensures that products with more similar attribute vectors are closer substitutes: the cross-product penalty for consuming products j and ℓ jointly is proportional to $\eta \mathbf{x}_j' \mathbf{x}_\ell$.

Demand. The consumer maximizes (17) subject to $q_0 + \mathbf{p}'\mathbf{q} \leq y$. Substituting the budget constraint, the first-order condition yields

$$\mathbf{q}(\mathbf{p}) = \frac{1}{\beta} (I_J + \eta XX')^{-1} (\boldsymbol{\alpha} - \mathbf{p}), \quad (18)$$

which is always well defined because $I_J + \eta XX'$ is positive definite.

The substitution matrix. Define the $J \times J$ matrix $\Theta := (\theta_{j\ell})_{j,\ell=1}^J$ with

$$\theta_{j\ell} := \mathbf{x}_j' \Omega^{-1} \mathbf{x}_\ell, \quad \Omega := \frac{1}{\eta} I_K + X'X. \quad (19)$$

It can be verified that Ω is a positive-definite weighting matrix, and $\theta_{j\ell} \in (-1, 1)$ for $j \neq \ell$ and $\theta_{jj} \in (0, 1)$.¹⁸

¹⁷[Antón et al. \(2023\)](#) use diversion ratios from [Pellegrino \(2025\)](#) to show that mergers increasingly occur between firms that sell similar products.

¹⁸Because $\eta > 0$, the matrix $\Omega = \frac{1}{\eta} I_K + X'X$ is positive definite: for any nonzero $\mathbf{u} \in \mathbb{R}^K$,

Proposition 4 *The demand system can be written in matrix form as*

$$\mathbf{q}(\mathbf{p}) = \frac{1}{\beta}(I_J - \Theta)(\boldsymbol{\alpha} - \mathbf{p}). \quad (20)$$

Equivalently, for each product j ,

$$q_j(\mathbf{p}) = a_j - \frac{1}{\beta}(1 - \theta_{jj})p_j + \frac{1}{\beta} \sum_{\ell \neq j} \theta_{j\ell} p_\ell,$$

where a_j is a product-specific demand intercept given by $\mathbf{a} = \frac{1}{\beta}(I_J - \Theta)\boldsymbol{\alpha}$.

Because preferences are quadratic, demand is linear in prices. All substitution patterns are encoded in the $J \times J$ matrix Θ with entries $\theta_{j\ell}$. The Slutsky matrix (Jacobian of demand) takes the form $S = -(I_J - \Theta)/\beta$. Products j and ℓ are substitutes when $\theta_{j\ell} > 0$ and complements when $\theta_{j\ell} < 0$. Crucially, $\theta_{j\ell}$ is a *weighted inner product* of the attribute vectors $\mathbf{x}_j$ and $\mathbf{x}_\ell$, which means that the closer j and ℓ are in this inner-product space, the more substitutable they are.

The next result shows that the weights in the inner product $\theta_{j\ell}$ have a transparent interpretation in terms of the principal components of the characteristics matrix X .

Proposition 5 *Let U be the $K \times K$ matrix whose columns are the principal-component directions of X , and let $\tilde{\mathbf{x}}_j = U'\mathbf{x}_j$ be the projection of $\mathbf{x}_j$ onto these directions. Then*

$$\theta_{j\ell} = \sum_{k=1}^K \frac{\tilde{x}_{jk} \tilde{x}_{\ell k}}{1/\eta + \lambda_k^{X'X}}, \quad (21)$$

where $\lambda_k^{X'X}$ is the k -th eigenvalue of $X'X$.

Expression (21) delivers two insights. First, because U is orthogonal, the rotation from $\mathbf{x}_j$ to $\tilde{\mathbf{x}}_j$ does not alter the inner-product structure ($\tilde{X}\tilde{X}' = XX'$), so the projection is without loss.

Second, the weight on the k -th principal component is inversely related to the variance of that component across all products. To see this, note that $\lambda_k^{X'X} = \tilde{\mathbf{x}}_k' \tilde{\mathbf{x}}_k$, so characteristics along which products are more homogeneous receive a higher weight. The intuition is that attributes shared by nearly all products in the market are especially informative about pairwise substitutability: consumers cannot easily switch to a differentiated alternative along a dimension on which all options are similar. Conversely, attributes that vary widely across products matter less for any given pair.

$\mathbf{u}'\Omega\mathbf{u} = \frac{1}{\eta}\|\mathbf{u}\|^2 + \|X\mathbf{u}\|^2 > 0$. Moreover, $\Theta = X\Omega^{-1}X'$ is the ridge *hat matrix*. Writing an SVD $X = U\Sigma V'$, we have $\Theta = U\Sigma(\Sigma^2 + \frac{1}{\eta}I)^{-1}\Sigma U'$, so its eigenvalues are $\sigma_r^2/(\sigma_r^2 + 1/\eta) \in [0, 1)$, which implies $\theta_{jj} = \mathbf{e}_j'\Theta\mathbf{e}_j \in (0, 1)$ (provided $\mathbf{x}_j \neq \mathbf{0}$). Finally, since $\Omega^{-1} \succ 0$, letting $\mathbf{z}_j := \Omega^{-1/2}\mathbf{x}_j$ gives $\theta_{j\ell} = \mathbf{z}_j'\mathbf{z}_\ell$, and thus by Cauchy-Schwarz, $|\theta_{j\ell}| \leq \|\mathbf{z}_j\| \|\mathbf{z}_\ell\| = \sqrt{\theta_{jj}\theta_{\ell\ell}} < 1$ for $j \neq \ell$.

To sum up, the substitutability between products j and ℓ depends not only on how similar their attribute vectors are, but also on how the similarity is distributed across dimensions relative to market-wide variation.

4.2 Product Attributes, Diversion Network, and Merger Impacts

We now connect the substitution matrix Θ to the diversion ratio matrix D introduced in Section 2. Recall that D is defined as $D = I_J - [\text{diag}(S)]^{-1}S$, so its off-diagonal entries are

$$D_{j\ell} = -\frac{S_{j\ell}}{S_{jj}} = \frac{\theta_{j\ell}}{1 - \theta_{jj}}, \quad j \neq \ell.$$

Under the linear-quadratic demand model, the diversion ratio from product j to product ℓ is therefore determined entirely by the attribute-based similarity $\theta_{j\ell}$, normalized by the own-price sensitivity term $1 - \theta_{jj}$. In particular:

- $D_{j\ell}$ is large when $\mathbf{x}_j$ and $\mathbf{x}_\ell$ are similar in the Ω^{-1} -weighted inner product, i.e., when the two products are close substitutes in characteristic space.
- Like the linear and logit models discussed in Conlon and Mortimer (2021), the network model delivers diversion ratios that do not vary with price. However, the economic source of this constancy is entirely different. In our model, diversion is constant because substitution patterns are generated by the geometry of product attributes under quadratic preferences, not because of IIA or because treatment effects are homogeneous. This structure allows us to preserve realistic heterogeneity in demand intercepts and marginal utilities while obtaining closed-form markups and centrality-based merger effects.

The entire diversion matrix D can be constructed from product characteristics alone, without estimating a demand system. This has an immediate implication for the results of Sections 2–3.¹⁹ Since equilibrium markups are determined by the Bonacich centrality of the diversion network (Lemma 1 and 1'), and since merger impacts propagate through walks in this same network (Propositions 1–2 and their primed extensions), the entire centrality-based apparatus can be computed from product attributes.

The results of this Section, combined with those of Sections 2–3, deliver a practical procedure for antitrust screening:

Product characteristics $X \rightarrow$ Diversion network $D \rightarrow$ Centrality-based merger effects.

¹⁹We assume that a unique equilibrium, where all prices p_j lie in the interior of $[c_j, \alpha_j]$, exists for the linear-quadratic utility model throughout. A sufficient condition on the primitives, as shown in Appendix B, is $(1 - \theta_{jj})(\alpha_j - c_j) > \sum_{k \neq j} |\theta_{jk}|(\alpha_k - c_k)$ for all j .

We close this section by reiterating several properties that make this procedure attractive for policy.

Low data requirements and compatibility with modern data sources. The diversion matrix D is constructed from the attribute matrix X and the single parameter η (equation (19)). No price or quantity data, no estimation of a structural demand system, and no market-share information are needed to build the competition network. In settings where product characteristics are publicly available—as they are in automobiles, consumer electronics, pharmaceuticals, and many other industries—the entire screening exercise can be carried out with minimal data collection. At the same time, the framework places essentially no restriction on *how* characteristics are measured: recent advances in machine learning show that embeddings derived from text descriptions, images, or other unstructured product data can capture demand-relevant positioning and substitute for hand-coded attributes (Bajari et al., 2023; Compiani et al., 2025). Relatedly, Magnolfi et al. (2022) demonstrate that embeddings from perceived-distance surveys yield elasticity estimates comparable to those from observable attributes. Because our construction depends on X only through inner products, any representation of product attributes that preserves the same inner-product structure generates the same substitution patterns and thus the same diversion network and centrality-based screening measures.

No ex ante market definition. As emphasized by Hoberg and Phillips (2016), a key advantage of characteristic-based approaches is that they bypass the need to define a relevant market. In our framework, the network structure *is* the market: products that are close in attribute space share strong links and exert competitive pressure on each other, whereas distant products are effectively independent. Even a very broad product universe does not distort the analysis, because the inner-product weights in (21) automatically downweight characteristics that vary widely and hence are less informative about pairwise substitutability.

Micro-founded in a setting with differentiated products. Related to the previous point, the centrality measure naturally emerges in the equilibrium of a price-setting game between arbitrarily differentiated products. By contrast, while functions of merging parties’ shares related to ΔHHI are shown to be indicative of merger impacts in differentiated price competition for aggregative games (Nocke and Whinston, 2022), the exact micro-foundation for ΔHHI is obtained within a homogeneous-good Cournot competition framework.

Coherent under arbitrary ownership structure. A further practical advantage is that the centrality statistic used for screening is well defined under *any* ownership or control structure. Ownership enters the model as a primitive of the firms’ objective functions through the decision-weight matrix H . As shown in Sections 2–3, partial

acquisitions and common ownership modify incentives directly via the internalized diversion network D_H , which governs equilibrium pricing. The screening statistic is computed from this object, so changes in control—whether full mergers or minority stake acquisitions by incumbents or financial investors—affect the measure through the same structure that determines markups.

Scale-free substitution patterns. Under Bertrand competition, the elements $\theta_{j\ell}$ lie in $(-1, 1)$ regardless of the units in which characteristics are measured. This means that the diversion network—and hence the centrality measures and merger effects—are invariant to rescaling of the attribute vectors. In practice, no normalization of product characteristics is required.

5 Application to the Automobile Industry

In this section, we estimate the model using data on the European automobile industry. The same data have been used extensively in the empirical industrial organization literature (e.g., [Brenkers and Verboven \(2006\)](#)). We then introduce our empirical framework and use it to estimate demand, comparing the results with those obtained by estimating demand using discrete-choice models. Lastly, we simulate the effects of mergers under different demand models and assess the effectiveness of merger screening using centrality-based rules.

5.1 Data

We use publicly available data from [Brenkers and Verboven \(2006\)](#).²⁰ The dataset covers prices, quantities, and detailed product characteristics for virtually all passenger car models sold in five major European markets—Belgium, France, Germany, Italy, and the United Kingdom—between 1970 and 1999. Over this period, roughly 350 distinct models appear in the data, though many represent successive generations of existing nameplates (e.g., Volkswagen Golf, Toyota Corolla, BMW 5-Series). In total, the panel contains 11,549 model–market–year observations, corresponding to an average of about 80 models available per country in a given year.

For each model, we observe the list price for the base version offered in each market, as reported in contemporary consumer catalogs. Sales are measured using new vehicle registrations at the model range level. The dataset further includes a rich set of physical and performance characteristics drawn from the same catalogs. These variables capture vehicle size (e.g., weight, length, width, height), engine specifications

²⁰The database is available at: <https://sites.google.com/site/frankverbo/data-and-software/data-set-on-the-european-car-market?authuser=0> and has also been used in [Goldberg and Verboven \(2001\)](#) and [Goldberg and Verboven \(2005\)](#).

Table 1: Summary Statistics on the European Automobile Industry

Variable	Mean	SD	Bel	Fra	Ger	Ita	U.K.
Sales (units)	19813.24	37719.92	3925.42	23305.81	31002.55	24292.14	19784.34
Horsepower (kW)	57.14	23.88	56.53	56.17	57.45	57.22	58.42
Fuel inefficiency (liter/100 km)	8.18	1.72	8.22	8.12	8.24	8.08	8.22
Width (cm)	164.38	9.62	164.16	164.28	164.57	164.06	164.82
Height (cm)	140.43	4.62	140.34	140.46	140.49	140.63	140.27
Foreign (0-1)	0.81	0.39	1.00	0.75	0.73	0.76	0.77
Price Eur	8352.52	5540.92	7674.24	8315.98	8119.62	8378.81	9384.34
Price Eur (post-tax)	10089.23	6606.51	9439.35	10476.46	9283.47	9946.35	11388.33
Price Eur (Deflated)	10793.17	5962.59	9029.31	10944.30	8394.99	13292.90	12870.79
Subcompact (0-1)	0.28	0.45	0.28	0.28	0.28	0.31	0.26
Compact (0-1)	0.23	0.42	0.23	0.24	0.23	0.22	0.24
Intermediate (0-1)	0.22	0.41	0.22	0.22	0.22	0.21	0.23
Standard (0-1)	0.19	0.39	0.19	0.19	0.18	0.17	0.19
Luxury (0-1)	0.08	0.27	0.08	0.07	0.09	0.09	0.08

Notes: The number of observations is 11549; SD means standard deviation.

(horsepower and displacement), and performance metrics (top speed, acceleration, and fuel economy). Finally, we observe identifiers for the model, brand, and manufacturer, along with the country of origin or production location and the market segment classification (subcompact, compact, intermediate, standard, luxury).

Table 1 provides summary statistics for all the key characteristics of the cars in our sample across market and years, and separately for each country across years. The last five rows show the penetration of each segment: not surprisingly, the luxury segment only accounts for about 8% of the sales. By contrast, the subcompact segment tends to be the largest (28% of the sales), followed closely by the compact and intermediate segments, with 23% and 22% of the sales, respectively.

5.2 The Empirical Nash-Bertrand Network Model

We operationalize the network model described in Section 4, showing how to adapt it to the specific context of the automobile industry.

An implicit assumption of our model is the absence of income effects due to the quasi-linearity of consumer preferences in the outside good. In the context of car purchases, this assumption might be unrealistic. For this reason, rather than parsimoniously extending the demand model to accommodate income effects, we follow [Berry, Levinsohn, and Pakes \(1995\)](#) and model consumer preference for the inside and outside goods in a Cobb-Douglas fashion:

$$U(y - \mathbf{p}'\mathbf{q}, \mathbf{q}; X) = (y - \mathbf{p}'\mathbf{q})^\gamma [G(\mathbf{q}, X)]^\phi \quad (22)$$

where the first term in U already substitutes for the budget constraint and

$$G(\mathbf{q}, X) \equiv \exp \left\{ \mathbf{q}' \boldsymbol{\alpha} - \frac{\beta}{2} \mathbf{q}' (I_J + \eta X X') \mathbf{q} \right\}.$$

Next, substituting G into (22) and taking logs, the representative consumer's utility can be written as

$$u(\mathbf{q}, X) \equiv \log(U) = \gamma \log(y - \mathbf{p}' \mathbf{q}) + \mathbf{q}' \boldsymbol{\alpha} - \frac{\beta}{2} \mathbf{q}' (I_J + \eta X X') \mathbf{q} \quad (23)$$

$$\approx \mathbf{q}' \left(\boldsymbol{\alpha} - \frac{\mathbf{p}}{y} \right) - \frac{\beta}{2} \mathbf{q}' (I_J + \eta X X') \mathbf{q} \quad (24)$$

where we normalize ϕ and γ to 1 because they cannot be separately identified from α and β , and we use a first-order Taylor expansion to approximate $\log(y - \mathbf{p}' \mathbf{q})$.²¹ We also calibrate η to 0.12 following Pellegrino (2025). The preferences in (24) are identical to the ones described in Section 4 except for the fact that prices are now measured relative to income. The representative consumer's demand system can be derived as before:

$$\mathbf{q}(\mathbf{p}) = \frac{1}{\beta} (I_J + \eta X X')^{-1} \left(\boldsymbol{\alpha} - \frac{\mathbf{p}}{y} \right). \quad (25)$$

To map this framework to automobile data, let t denote the market (i.e., a country-year pair) and note that equation (25) can be rearranged in vector form as

$$\tilde{\mathbf{q}}_t \equiv (I_{J_t} + \eta X_t X_t') \mathbf{q}_t = \frac{1}{\beta} \left(\boldsymbol{\alpha}_t - \frac{\mathbf{p}_t}{y_t} \right) \quad (26)$$

where J_t is the number of car models available in market t , X_t is a $J_t \times K$ matrix of product characteristics and the linear preference parameter vector α is allowed to vary over time. Next, consider the j th equation of the above system

$$\tilde{q}_{jt} = \frac{1}{\beta} \left(\alpha_{jt} - \frac{p_{jt}}{y_t} \right) \quad (27)$$

and note that, upon calibrating η and assuming that the matrix X_t contains only observable characteristics, the left-hand side in (25), denoted by $\tilde{q}_{jt}$, is directly measurable. Conversely, on the right-hand side of (25), only $\mathbf{p}_t$ and y_t are observable.²²

More generally, equation (27) suggests that we can estimate demand using the

²¹The same approximation has been used in Berry, Levinshon, and Pakes (1999).

²²To measure income in a given year, we use the simulated draws that come with the pyBLP package developed in Conlon and Gortmaker (2020), and we average them using the weights provided. The draws come from a log-normal distribution of income whose location and scale parameters are estimated from the Current Population Survey (CPS) each year as described in Berry, Levinsohn, and Pakes (1995).

following linear specification

$$\tilde{q}_{jt} = -\frac{1}{\beta} \frac{p_{jt}}{y_t} + w'_{jt}\zeta + \xi_{jt} \quad (28)$$

where w_{jt} is a vector of observable product and demographic characteristics which, in this context, includes both x_{jt} and y_t . On the other hand, ξ_{jt} includes characteristics that are unobservable to the econometrician but known by the agents.

To sum up, two assumptions allow us to estimate demand from (28). First, all the characteristics that enter consumer preferences in the quadratic term are observable. Second, any unobserved characteristic (ξ_{jt}) enters consumer preferences only through the linear parameter vector α , i.e.,²³

$$\alpha_{jt} = \beta(w'_{jt}\zeta + \xi_{jt}) \quad (29)$$

To consistently estimate (28), we need to instrument p_{jt}/y_t because prices will be correlated with the unobservable component ξ_{jt} . The reason is that firms internalize ξ_{jt} before setting prices simultaneously. To see this formally, recall that Nash equilibrium prices are given by

$$p_{jt} = c_{jt} + \frac{y_t \alpha_{jt} - c_{jt}}{2} - b_{jt} \quad (30)$$

and note that those prices are a function of α_{jt} (and in turn function of ξ_{jt}) both directly and indirectly through product j 's Bonacich centrality b_{jt} . Under this setting, demand can be estimated with a simple linear instrumental variable strategy which is described next.

5.3 Demand Estimation

We begin by estimating demand from the linear specification in (28). To do so, we instrument the term p_{jt}/y_t using the set of demand instruments z_{jt} constructed in [Brenkers and Verboven \(2006\)](#). These include the product's own observed characteristics (horsepower, fuel efficiency, width, height, and a dummy for whether the product is a foreign one), as well as functions of competitors' characteristics ([Berry et al., 1995](#)). Specifically, we use: the number of products and the sums of characteristics of other products of the same firm belonging to the same segment, and the number of products and the sums of the characteristics of competing products belonging to the same segment.

Equation (30) implies product characteristics affect firms' pricing decisions through

²³An additional assumption is that β is constant across markets. However, this can be partially relaxed by interacting prices with any market level observable in equation (28).

Table 2: Estimated Substitution Patterns across Demand Models

Segment	Own-price	Cross-price elasticity		Diversion ratio	
		Within segment	Across segments	Within segment	Across segments
Logit					
Subcompact	-1.32	0.00	0.00	0.05	0.05
Compact	-1.85	0.00	0.00	0.05	0.05
Intermediate	-2.19	0.00	0.00	0.04	0.04
Standard	-3.06	0.00	0.00	0.02	0.02
Luxury	-4.22	0.00	0.00	0.03	0.03
Nested Logit					
Subcompact	-2.25	0.03	0.00	1.30	0.05
Compact	-3.08	0.05	0.00	1.51	0.04
Intermediate	-2.77	0.00	0.00	0.04	0.03
Standard	-3.96	0.01	0.00	0.15	0.02
Luxury	-6.99	0.40	0.00	6.12	0.02
RC Logit					
Subcompact	-1.54	0.02	0.02	1.12	0.97
Compact	-2.15	0.02	0.02	1.06	1.06
Intermediate	-2.55	0.02	0.02	0.86	0.82
Standard	-3.59	0.02	0.02	0.63	0.47
Luxury	-4.93	0.04	0.02	1.03	0.59
RC Nested Logit					
Subcompact	-2.87	0.07	0.02	2.33	0.82
Compact	-3.47	0.07	0.02	2.20	0.84
Intermediate	-3.15	0.03	0.02	0.82	0.60
Standard	-4.55	0.03	0.02	0.78	0.36
Luxury	-7.72	0.44	0.02	6.19	0.48
Network					
Subcompact	-4.50	0.11	0.06	2.31	0.73
Compact	-5.46	0.13	0.05	2.35	0.82
Intermediate	-6.93	0.17	0.06	2.38	0.85
Standard	-9.53	0.24	0.06	2.53	0.87
Luxury	-9.33	0.29	0.05	3.29	0.96

Notes: The table summarizes, for each of the five demand models considered, the average elasticities and diversion ratios across all markets by segment. Values are trimmed at the 25th percentile for the Network model.

the centrality term b_{jt} and the observable part of α_{jt} , suggesting the relevance of our instrument. To ensure that demand remains constant while instruments shift the supply, the identifying assumption relies on the idea that firms choose characteristics before observing any demand shock ξ_{jt} which formally boils down to requiring that $\mathbb{E}[\xi_{jt}|z_{jt}] = 0$.

Because θ_{jl} is a function of observable product characteristics, estimating $1/\beta$ is sufficient to recover own-price elasticities, cross-price elasticities, and diversion ratios. The bottom part of Table 2 shows the estimates we obtain for the average diversion ratio, own- and cross-price elasticities, both within and across segments. For example, our estimates imply that following a marginal increase in the price of a compact car, on

average 2.35% of the lost sales are diverted to other compact models, whereas 0.82% are diverted to models in other segments.

Comparison with discrete-choice models. We compare the network demand estimates with four standard discrete-choice specifications: Logit, Nested Logit, Random Coefficients Logit (RCL), and Random Coefficients Nested Logit (RCNL). All models are estimated using the same product-level data and the same set of observable characteristics. In each case, price is treated as endogenous and instrumented using the BLP-style instruments described above.

We first estimate a homogeneous multinomial logit model. Consumer utility is specified as

$$u_{ijt} = x_{jt}\beta - \alpha \frac{p_{jt}}{y_t} + \xi_{jt} + \varepsilon_{ijt}, \quad (31)$$

where x_{jt} includes observable characteristics and ε_{ijt} follows the type-I extreme value distribution. Mean utility is recovered using the standard logit inversion

$$\delta_{jt} = \log s_{jt} - \log s_{0t}. \quad (32)$$

We then estimate

$$\delta_{jt} = x_{jt}\beta - \alpha \frac{p_{jt}}{y_t} + \xi_{jt} \quad (33)$$

via two-stage least squares. This specification imposes homogeneous price sensitivity and the Independence of Irrelevant Alternatives (IIA), and hence substitution patterns depend only on market shares.

To relax the IIA property, we estimate a nested logit model that groups vehicles into the five market segments available in our data: subcompact, compact, intermediate, standard, and luxury. This allows consumers to have correlated preferences for products in the same segment, which share a common set of features. Utility is specified as:

$$u_{ijt} = \delta_{jt} + \zeta_{ig} + (1 - \sigma)\varepsilon_{ijt}, \quad (34)$$

where g indexes the nest and $\sigma \in [0, 1)$ captures within-segment correlation. The estimating equation becomes

$$\log s_{jt} - \log s_{0t} = x_{jt}\beta - \alpha \frac{p_{jt}}{y_t} + \sigma \log s_{j|g,t} + \xi_{jt}, \quad (35)$$

where $s_{j|g,t}$ denotes the within-nest share. Both price and the within-nest share term

are instrumented using BLP-style excluded instruments. Specifically, we interact these instruments with the set of segment dummies to allow the nesting parameter to differ across segments. This model allows stronger substitution within segments than across segments while maintaining homogeneous price sensitivity across consumers.

We also estimate an RCL model in the spirit of [Berry, Levinsohn, and Pakes \(1995\)](#). Utility is specified as

$$u_{ijt} = x_{jt}\beta - \alpha \frac{p_{jt}}{y_t} + \mu_{ijt} + \xi_{jt} + \varepsilon_{ijt}, \quad (36)$$

where μ_{ijt} captures unobserved taste heterogeneity through random coefficients on selected product characteristics and a random intercept. Specifically, coefficients on observable characteristics are assumed to vary across consumers according to a multivariate normal distribution.

Importantly, the price coefficient remains homogeneous across consumers. We do not model income as a demographic interaction with price in these discrete-choice specifications because we do not observe the income distribution at the market-year level for the European automobile markets in our sample. The model is estimated by simulated GMM using the `pyBLP` package, integrating over the distribution of random coefficients ([Conlon and Gortmaker, 2020](#)). Price is instrumented using the same excluded instruments as in the logit specification.

Finally, we estimate an RCNL model, which combines taste heterogeneity with within-segment correlation ([Grigolon and Verboven, 2014](#)). Utility takes the form:

$$u_{ijt} = x_{jt}\beta - \alpha \frac{p_{jt}}{y_t} + \mu_{ijt} + \sigma\zeta_{ig} + \xi_{jt} + \varepsilon_{ijt}, \quad (37)$$

where μ_{ijt} represents random coefficients on product characteristics (including a random intercept) and σ captures within-nest correlation. As in the RCL specification, price sensitivity is not heterogeneous across consumers. The model is estimated via simulated GMM using `pyBLP`, with the same instrument set used in the previous specifications.

Across models, substitution patterns become progressively more flexible. Logit imposes IIA and homogeneous price sensitivity; Nested Logit allows stronger substitution within segments; RCL introduces unobserved taste heterogeneity; and RCNL combines taste heterogeneity with within-segment correlation. These differences clearly emerge from Table 2.²⁴ In Logit and RCL, average diversion ratios are similar within and across segments, reflecting the absence of explicit segment-level correlation. In contrast, Nested Logit and RCNL generate higher within-segment diversion, as preference

²⁴Table C.1 in the appendix shows the results we obtain when the values of the own-price elasticities are trimmed at the 10th percentile before averaging.

correlation mechanically strengthens substitution among products in the same nest. Interestingly, the network model is able to replicate this pattern without exogenously imposing a nesting structure, generating average diversion ratios that are remarkably close to those of the RCNL specification. This is particularly valuable because, in many settings, it is difficult to determine the appropriate nesting structure *ex ante*, and mis-specifying the nests can lead to biased estimates of substitution patterns (Fosgerau et al., 2024).

We also emphasize that the network model allows consumers to exhibit a taste for variety and to consume more than one good. In this respect, the automobile industry—where households typically own no more than two cars and, in most cases, purchase only one vehicle at a time—does not represent the ideal environment to apply our framework. Therefore, the fact that the model performs well in this setting provides a conservative assessment of its empirical relevance. Its performance is likely to be even stronger in industries characterized by multi-product consumption and richer substitution patterns.

5.4 Merger Simulations

Next, we consider all possible mergers—across all countries and years—between the car manufacturers in our sample. In what follows, we base our analyses on demand estimates obtained from the network and RCNL models.

The first step entails recovering firms’ marginal costs. To that end, we estimate $\hat{\alpha}_{jt}$ for each product j and market t by simply plugging our demand estimates ($\hat{\beta}, \hat{\zeta}$) and the estimated regression residuals ($\hat{\xi}_{jt}$) into equation (29). Then, from the pricing equation in (30), we can recover marginal costs c_{jt} , price-cost margins $p_{jt} - c_{jt}$ and decompose the markup into monopolistic price-cost margins $(\alpha_{jt} - c_{jt})/2$ and network centrality b_{jt} . This decomposition implies that, in the network model, network centrality is the primary determinant of the effects of acquisitions.

The median marginal cost across models and markets is \$7,636.17 (\$6,493.51), and 75% of car models have a marginal cost lower than \$12,555.56 (\$10,839.04) in the network (RCNL) framework.²⁵ Overall, the estimated marginal costs are reasonably close across the two demand models considered.

Figure 1 compares merger simulations obtained from the network model and the RCNL specification, holding marginal costs fixed at pre-merger levels. The two models generate strongly correlated predictions across all outcomes, indicating substantial agreement in the ranking of mergers by their competitive impact. This is consistent with the broadly similar diversion patterns implied by the two demand systems. The

²⁵Figure C.1 shows the distribution of marginal costs across the network and the RCNL models, focusing on the most recent year of our sample (1999).

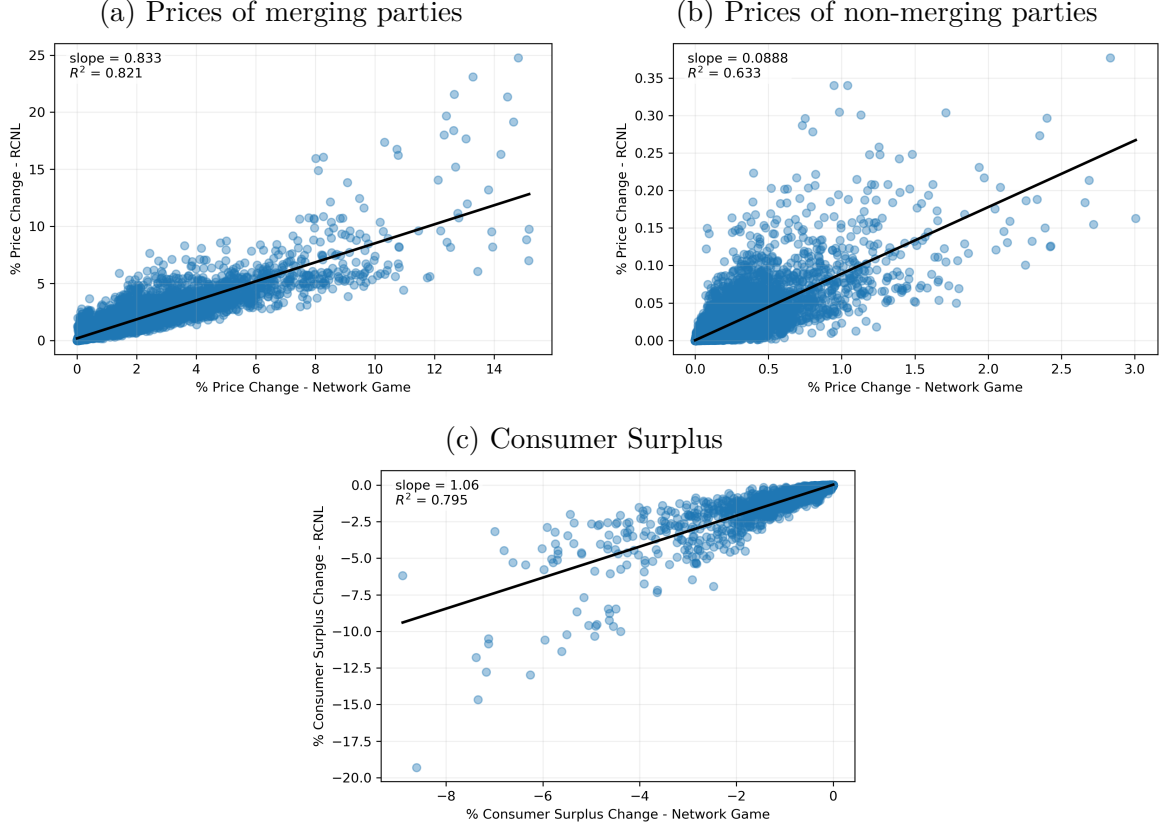


Figure 1: Simulation of the effects of all possible mergers

Notes: Each figure contrasts the effect of a potential merger estimated using the network model with that estimated using the RCNL model. Each figure focuses on the percentage change in a different outcome: the prices of the merged parties, the prices of the non-merged parties, and consumer surplus.

plots, however, also reveal some systematic differences. The network model tends to predict smaller price increases and smaller consumer-surplus losses, consistent with its higher (in absolute value) own-price elasticities. At the same time, merger outsiders appear more responsive under the network specification: non-merging products experience relatively larger price adjustments compared to RCNL, suggesting that competitive effects propagate more broadly through the similarity network rather than remaining concentrated within predefined nests.

To gain a better understanding of these differences in magnitudes, we zoom in on domestic mergers, which are arguably more likely to raise competitive concerns, and compare the simulation results obtained by relying on demand estimates obtained using the RCNL and the network models. To tag domestic mergers, we use the information on the country of origin or production location of each manufacturer. Since some manufacturers produce in multiple countries (e.g., BMW and Ford), simulated mergers involving them can span multiple countries. Table 3 summarizes the results for the potential domestic mergers that could have happened in 1999.

We find that the two models generate reasonably close price and CS effects across all

Table 3: Domestic Mergers in 1999: RCNL vs. Network

Country	Merger	ΔHHI	$\Delta P\%$ - Merged		$\Delta P\%$ - Non-Merged		$\Delta CS\%$	
			RCNL	Network	RCNL	Network	RCNL	Network
FRA	Peugeot–Renault	1623.5437	15.9407%	8.0120%	0.3048%	0.9834%	-5.4262%	-3.3852%
GBR	BMW–Ford	348.1630	3.7497%	2.8420%	0.0418%	0.3665%	-1.1106%	-0.8891%
GBR	BMW–Opel	255.0992	2.9597%	2.2277%	0.0455%	0.2677%	-0.8305%	-0.6477%
GBR	Ford–Opel	651.7778	5.3994%	3.5448%	0.0782%	0.4839%	-2.7334%	-1.5025%
GER	BMW–Ford	151.0954	1.9720%	2.1863%	0.0343%	0.2974%	-0.3921%	-0.3608%
GER	BMW–MCC	18.7394	0.3090%	0.4900%	0.0026%	0.0260%	-0.0236%	-0.0297%
GER	BMW–Mercedes	155.4387	3.0930%	2.0623%	0.0768%	0.2260%	-0.9659%	-0.3604%
GER	BMW–Opel	234.0387	2.3207%	2.6840%	0.0438%	0.3396%	-0.5545%	-0.4789%
GER	BMW–VW	494.3758	3.9663%	4.4607%	0.0655%	0.8168%	-1.4288%	-1.1102%
GER	Ford–MCC	24.0961	0.9525%	0.7483%	0.0089%	0.0408%	-0.0762%	-0.0516%
GER	Ford–Mercedes	199.8714	2.9257%	2.3549%	0.0523%	0.2588%	-0.5967%	-0.4130%
GER	Ford–Opel	300.9395	3.6512%	3.3459%	0.0794%	0.4110%	-0.9313%	-0.6367%
GER	Ford–VW	635.6948	5.9811%	5.8639%	0.0768%	1.0172%	-1.9570%	-1.4532%
GER	MCC–Mercedes	24.7888	0.6152%	0.9370%	0.0033%	0.0233%	-0.0303%	-0.0359%
GER	MCC–Opel	37.3236	1.6758%	1.1744%	0.0139%	0.0474%	-0.1265%	-0.0711%
GER	MCC–VW	78.8412	1.5702%	1.3616%	0.0120%	0.1153%	-0.2077%	-0.1548%
GER	Mercedes–Opel	309.5901	3.1861%	2.3994%	0.0684%	0.2526%	-0.8399%	-0.5392%
GER	Mercedes–VW	653.9680	4.0745%	3.6063%	0.0948%	0.6038%	-2.0943%	-1.2339%
GER	Opel–VW	984.6571	6.3387%	6.1152%	0.1091%	1.0259%	-3.0836%	-1.9051%

Notes: The table reports simulated effects of hypothetical domestic mergers among automobile manufacturers in 1999. ΔHHI denotes the change in the HHI resulting from the merger, computed using pre-merger market shares and expressed on the 0–10,000 scale. Columns 4–5 report the average percentage price change for the products of the merging firms, while Columns 6–7 report the average percentage price change for the products of non-merging firms. Columns 8–9 report the percentage change in consumer surplus relative to the pre-merger equilibrium. Results are computed using the RCNL model and the network demand model described in the paper. All price and welfare changes are computed from the post-merger equilibrium implied by each model, holding marginal costs fixed at the pre-merger level.

simulated domestic mergers. For example, a merger between the two largest domestic manufacturers in France (Peugeot and Renault), which raises the HHI by almost 1,624 points, leads the merged entity to increase prices by 8.0% under the network model and by 16% under RCNL. The associated consumer-surplus loss amounts to 3.4% in the network model and 5.4% in RCNL. By contrast, a merger between BMW and MCC in Germany, where the firms have limited overlap across segments, induces a small change in HHI (less than 19 points) and generates much smaller effects under both models, with price increases between 0.3–0.49% and CS losses between 0.02–0.03%.

On average, the network model predicts more moderate price effects than the RCNL model, especially for the merging parties. This difference is potentially relevant in light of recent evidence that merger simulations based on random-coefficients demand models can substantially overestimate realized post-merger price increases (Illanes et al., 2025). At the same time, the comparison is not uniform across acquisitions. For instance, in the BMW–VW merger in Germany, the network model predicts larger price increases for both merged and non-merged products relative to RCNL, yet the associated CS loss is smaller. This may reflect differences in demand curvature and

substitution patterns across the two specifications. Although the network model implies stronger diversion between BMW and VW, leading to larger unilateral price effects, it also generates broader substitution toward alternative inside products and relatively less reallocation toward the outside option. As a result, market contraction is attenuated, and the overall welfare loss is smaller despite larger price increases.

5.5 Changes in Bonacich Centrality as a Screening Tool

Competition authorities have long relied on concentration indices (post-merger HHI and ΔHHI) to assess the competitive effects of horizontal acquisitions. The 2010 Merger Guidelines establish that markets with an HHI below 1,500 are “unconcentrated” whereas markets with an HHI above 2,500 are “highly concentrated.”²⁶ In “unconcentrated” markets, deals are unlikely to be challenged. In highly concentrated markets, mergers are unlikely to be challenged only when the ΔHHI is below 100. In contrast, mergers are likely to be challenged when the ΔHHI is greater than 200. The other mergers happening in highly concentrated markets are “more likely than not” to be challenged. In the remaining “moderately concentrated” markets, with post-merger HHI between 1,500 and 2,500, a merger is considered more likely than not to be challenged if ΔHHI is above 100; otherwise, agencies are unlikely to challenge it.

These thresholds have recently been updated by the 2023 Merger Guidelines, which have re-established, as per the 1982 Merger Guidelines, markets with an HHI greater than 1,800 as highly concentrated. Additionally, the new guidelines introduced a structural presumption: deals leading to a change in HHI greater than 100 in highly concentrated markets or involving parties with a combined market share above 30% are presumed to substantially lessen competition or tend to create a monopoly. We refer to this criterion as the 2023 concentration-based presumption (CBP).

Nocke and Whinston (2022) provides a foundation for using solely ΔHHI in screening mergers for whether their unilateral price effects will harm consumers. In our framework, the decomposition of the price-cost margin into a monopolist and a centrality component (Lemma 1') and the theoretical merger analysis (Proposition 3') provide a rationale for the Bonacich Centrality as a novel measure to be used for merger screening. Specifically, we define a market- or industry-level index, which we call *centrality*, as $\mathbf{q}'\mathbf{b}\left(D_H, \frac{1}{2}, \frac{\alpha-c}{2}\right)$ —i.e., the quantity-weighted sum of all products' Bonacich centrality.²⁷ As mentioned in Section 2, for non-linear demands, the internalized diversion ratio matrix D_H and demand intercept α are obtained through local

²⁶The 1982 Meger Guidelines already establish similar thresholds. The 2010 Guidelines maintained the structure but made the thresholds less stringent.

²⁷Appendix C.1 provides additional details on the computation of this index.

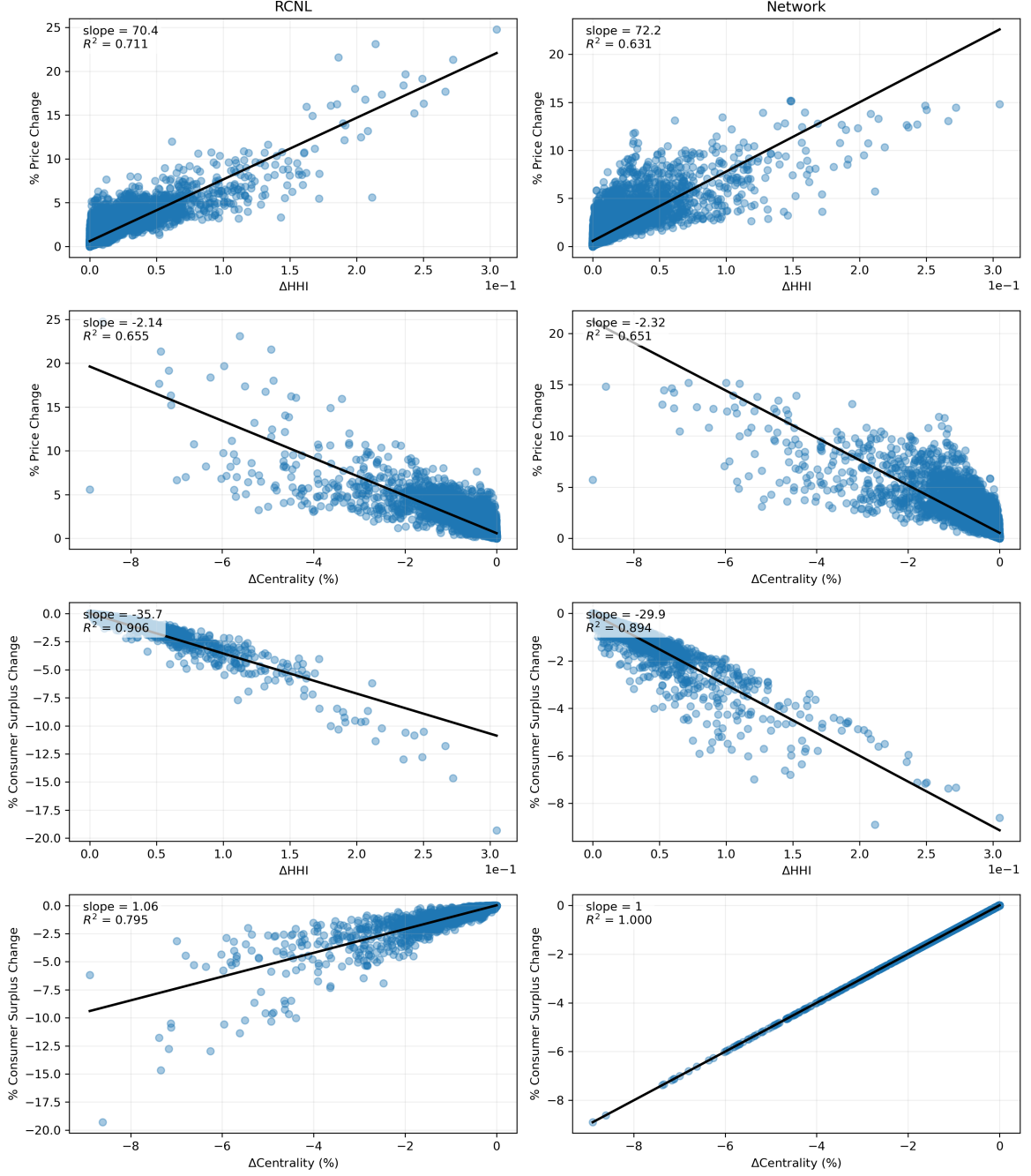


Figure 2: Correlation between the Effects of Simulated Mergers and Screening Tools

Notes: The figure shows the % price or consumer surplus changes against ΔHHI or $\Delta\text{Centrality}$ (in percentage changes), for all possible mergers across all markets (country-year). The figures on the left (right) use demand estimates from the RCNL (network) model.

linearization around the equilibrium prices.²⁸ Further, we define

$$\Delta\text{Centrality} := \mathbf{q}'\mathbf{b}\left(D_{H\text{pre-merger}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right) - \mathbf{q}'\mathbf{b}\left(D_{H\text{post-merger}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right).$$

²⁸When demand is estimated using prices normalized by income, the linearized system is $\mathbf{q} = -\tilde{S}(\tilde{\boldsymbol{\alpha}} - \mathbf{p}/y)$; to compute Bonacich centralities in the original price units, we then set $S = \tilde{S}/y$ and $\boldsymbol{\alpha} = y\tilde{\boldsymbol{\alpha}}$.

Table 4: Simulated Price Effects and Screening Tools

	(1) % ΔP	(2) % ΔP	(3) % ΔP	(4) % ΔP	(5) % ΔP	(6) % ΔP	(7) % ΔP
Panel A: RCNL							
ΔHHI	0.702*** (0.014)			0.533*** (0.022)	0.343*** (0.016)		0.071*** (0.020)
$\Delta \text{Centrality}$		-2.381*** (0.139)		-0.794*** (0.104)		-1.174*** (0.072)	-1.056*** (0.079)
GUPPI			1.032*** (0.016)		0.613*** (0.017)	0.749*** (0.018)	0.690*** (0.016)
N	21786	21786	21786	21786	21786	21786	21786
R^2	0.738	0.609	0.765	0.763	0.814	0.855	0.856
Panel B: Network game							
ΔHHI	0.717*** (0.016)			0.325*** (0.028)	0.604*** (0.026)		0.100*** (0.023)
$\Delta \text{Centrality}$		-2.803*** (0.157)		-1.835*** (0.187)		-2.128*** (0.156)	-1.963*** (0.172)
GUPPI			0.932*** (0.018)		0.193*** (0.027)	0.419*** (0.030)	0.336*** (0.025)
N	21786	21786	21786	21786	21786	21786	21786
R^2	0.670	0.735	0.543	0.784	0.677	0.802	0.804

Notes: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. The dependent variable is the estimated percentage price change for the merging parties following the merger. $\Delta \text{Centrality}$ and GUPPI are in percentage terms. All regressions include market (country-year) fixed effects. Standard errors in parentheses.

Next, we assess the performance of $\Delta \text{Centrality}$ in predicting the unilateral effects of mergers. We compare the performance with that of ΔHHI , and investigate to what extent the novel measure of $\Delta \text{Centrality}$ may complement traditional tools adopted in merger screening.²⁹ We emphasize that $\Delta \text{Centrality}$ is an industry-level index, much like conventional concentration measures. This is because a merger affects the centrality scores of all products, not only those of the merging parties. As a result, focusing only on the centrality change of the merging products would overlook the part of the merger incentives generated by network propagation.

Figure 2 shows the predicted percentage changes across all possible mergers in the prices of merging parties or CS against either $\Delta \text{Centrality}$ or ΔHHI .

The four left figures rely on estimates from the RCNL model, whereas the four right ones use the ones from the network model. Overall, across demand models, mergers leading to a larger decrease in centrality generate greater price increases and reductions in CS. In the network model, the relationship between CS losses and the percentage change in centrality is exact, as established in Section 3, and is therefore reflected in the bottom-right panel of Figure 2. Additionally, our analysis confirms

²⁹Consistent with the idea that concentration indices can be complemented with observable characteristics to help capture substitution patterns, Benson et al. (2024) show that, in the banking industry, introducing a threshold based on the proximity of merging firms' branch networks could improve merger screening.

the strong predictive power of ΔHHI , in line with evidence from prior studies in other industries (e.g., [Bhattacharya et al. \(2023\)](#)): across both demand models, a higher increase in market concentration is associated with greater price increases by merging parties, as well as larger losses in CS.

The evidence presented so far suggests that $\Delta\text{Centrality}$ can serve as a useful screening statistic for mergers, while also confirming that ΔHHI performs well on its own. A natural question, however, is whether, by capturing substitution patterns overlooked in concentration indices, $\Delta\text{Centrality}$ adds incremental information beyond ΔHHI . We also compare it with the Gross Upward Pricing Pressure Index (GUPPI), a standard first-order pricing-pressure measure. This comparison is particularly relevant because FOAs were developed precisely to incorporate diversion-based information about closeness of competition that concentration measures may miss. In contrast to GUPPI, however, $\Delta\text{Centrality}$ captures not only incentives arising from the diversion among merging products, but also how those incentives propagate through the broader network of competing non-merging products.³⁰

To assess whether $\Delta\text{Centrality}$ adds information beyond ΔHHI and GUPPI, we regress the simulated percentage price change of the merging parties across all possible mergers on ΔHHI , $\Delta\text{Centrality}$, and GUPPI, first separately and then jointly in alternative combinations, controlling for market and year fixed effects.

Panel A reports results using the RCNL demand estimates. Column (1) shows that a 0.01 increase in ΔHHI , corresponding to a 100-point increase on the conventional 0–10,000 HHI scale, is associated with approximately a 0.7 percentage-point increase in post-merger prices. Column (2) indicates that a one-percent increase in $\Delta\text{Centrality}$ predicts a 2.4 percentage-point decrease in prices, while Column (3) shows that a one-percent increase in GUPPI is associated with a 1.0 percentage-point increase in prices. When $\Delta\text{Centrality}$ is included jointly with ΔHHI in Column (4), both coefficients remain statistically and economically significant, and the R^2 rises to 0.763 from 0.738 in the univariate specification.

Importantly for our framework, Column (6) shows that $\Delta\text{Centrality}$ also adds substantial predictive information beyond GUPPI: the R^2 increases from 0.765 with GUPPI alone to 0.855 when the two measures are included together. Additionally, comparing Columns (5) and (6) indicates that complementing GUPPI with $\Delta\text{Centrality}$ rather than with ΔHHI substantially increases explanatory power, raising the R^2 from 0.814 to 0.855. This is consistent with changes in centrality implied by mergers, capturing the propagation of price effects across the network instead of simply merging parties’ incentives to raise prices. In the fully saturated specification in Column

³⁰Since GUPPI is typically defined at the product level, in Section C.2 of the Appendix, we describe how we construct our aggregate merger-level GUPPI measure building on [Conlon and Mortimer \(2013\)](#).

(7), all three measures remain statistically significant, and, relative to the specification with only ΔHHI and GUPPI , the addition of $\Delta\text{Centrality}$ increases the R^2 from 0.763 to 0.856. By contrast, the additional contribution of ΔHHI beyond GUPPI and $\Delta\text{Centrality}$ in explaining the variation in prices implied by the merger is negligible.

Panel B presents the same exercise using price effects simulated from the network model. In this case, $\Delta\text{Centrality}$ is the strongest standalone predictor: its R^2 is 0.735, compared with 0.670 for ΔHHI and 0.543 for GUPPI . This is unsurprising, since centrality is theoretically grounded in the network structure of demand that generates the simulated merger effects. Still, the joint specifications are informative. When $\Delta\text{Centrality}$ is combined with ΔHHI in Column (4), the R^2 rises to 0.784, and both measures remain statistically significant. Column (6) shows that $\Delta\text{Centrality}$ also contributes substantial information beyond GUPPI , increasing the R^2 from 0.543 to 0.802. The fully saturated specification in Column (7) delivers only a modest further improvement, with an R^2 of 0.804.

Overall, these results indicate that changes in centrality provide complementary predictive power that is not subsumed by either concentration measures or traditional first-order pricing-pressure tools. This distinction is central to our framework. While ΔHHI summarizes changes in market structure and GUPPI captures direct diverted-margin incentives, $\Delta\text{Centrality}$ reflects how merger-induced pricing pressure propagates through the broader network of differentiated products. The fact that centrality remains highly informative even in the RCNL simulations suggests that this network component captures economically relevant variation in merger effects beyond the information contained in standard screening statistics.

Unweighted centrality. One caveat on the analyses presented is that, in practice, merger screening requires tools that are easy to construct with limited data. Our theoretical measure of $\Delta\text{Centrality}$ requires estimating the full network model to recover $(\alpha - c)/2$, which may not always be feasible for enforcement agencies operating under data constraints. To address this concern, we construct a simpler measure that replaces $(\alpha - c)/2$ with a vector of ones 1_J . The resulting statistic corresponds to an unweighted Bonacich centrality given by:

$$\hat{b}\left(D_H, \frac{1}{2}, 1_J\right) = \left(I_J - \frac{1}{2}D_H\right)^{-1} \frac{1}{2}D_H 1_J, \quad (38)$$

where Θ is constructed using the same characteristics that enter indirect utility in [Brenkers and Verboven \(2006\)](#).³¹ This measure still encapsulates diversion information but only requires product characteristics, which are often available for both merging and non-merging parties. Moreover, advances in data collection and machine learning

³¹Recall that η is calibrated to 0.12.

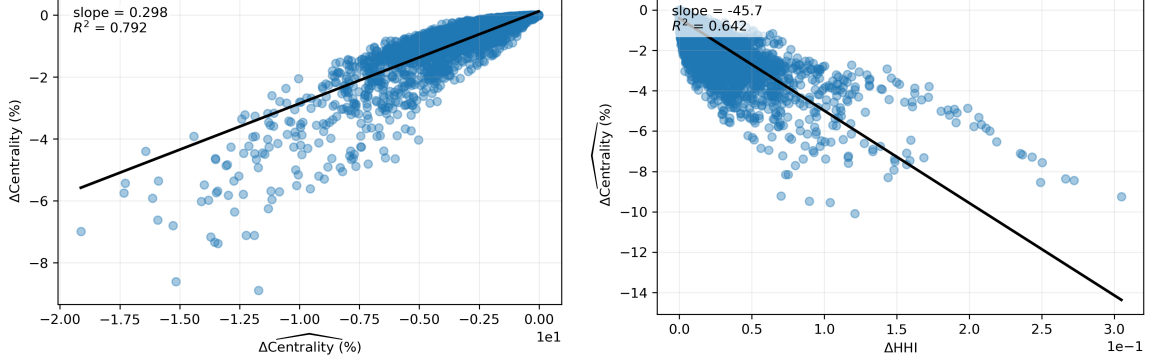


Figure 3: Unweighted Change in Centrality

Notes: The left (right) figure illustrates the relationship between the percentage change in unweighted centrality and $\Delta\text{Centrality}$ (ΔHHI) across all mergers.

tools can allow agencies to quickly extract such information from firm websites and industry sources.

For any potential merger, we then compute $\widehat{\Delta\text{Centrality}}$ as the change induced in (38). Figure 3 shows the strong correlation between the theoretical $\Delta\text{Centrality}$ and its unweighted counterpart, demonstrating that this simpler measure captures most of the economically relevant variation in our structural index. Moreover, we also show that $\widehat{\Delta\text{Centrality}}$ is highly correlated with the ΔHHI .

Performance in Merger Screening. We evaluate the performance of screening rules based on $\Delta\text{Centrality}$ against concentration-based presumptions and a rule that allows deals with a GUPPI lower than 5%. For screening rules using concentration indices, we consider the structural presumption introduced by the 2023 Merger Guidelines, as well as a rule solely based on the ΔHHI presumption that approves mergers with changes in concentration lower than 100.

To construct our centrality-based rules, we proceed as follows. Let $m = 1, \dots, M$ index potential mergers and $r \in \{\text{strict}, \text{baseline}, \text{lax}\}$ denote enforcement regimes, which differ in their assumed post-merger efficiency realizations. For each merger m and regime r , we compute (i) the change in consumer surplus $\Delta CS_{m,r}$ from a full merger simulation, (ii) the opposite of the change in centrality ΔC_m , and (iii) whether the merger would be privately profitable and therefore proposed, denoted by $1\{\Pi_{m,r} > 0\}$.³² The optimal screening rule only approves mergers that increase consumer surplus.³³

³²As before, the unit of analysis is a pair of firms in a given market (country-year). Thus, the same merger appears as a separate observation across markets.

³³The optimal screening rule is defined relative to consumer surplus effects simulated under the RCNL benchmark, rather than realized post-merger outcomes—which are unobserved. Recent evidence suggests that merger simulations based on random-coefficients demand models tend to over-predict realized post-merger price increases, while nonetheless often remaining more informative than

We restrict attention to monotone threshold rules based on ΔC_m , where this can be based on both the theoretical weighted centrality or the unweighted one. For a given cutoff τ , the regulator challenges the merger m if $\Delta C_m > \tau$ and allows it otherwise. For simplicity, we assume that challenged deals are blocked, though we recognize that, in practice, such mergers may still be allowed after further analysis. Let $d_m(\tau) = 1\{\Delta C_m > \tau\}$ denote the blocking decision. Conditional on the set of proposed mergers under regime r , $\mathcal{P}_r = \{m : \Pi_{m,r} > 0\}$, the regulator’s objective is to maximize expected consumer surplus from allowed mergers:

$$W_r(\tau) = \sum_{m \in \mathcal{P}_r} (1 - d_m(\tau)) \Delta C S_{m,r}. \quad (39)$$

The welfare-optimal Δ Centrality threshold in regime r is therefore

$$\tau_r^* = \arg \max_{\tau} W_r(\tau). \quad (40)$$

Because decisions change only when τ crosses an observed value of ΔC_m , the maximization can be performed by evaluating $W_r(\tau)$ over the finite set of candidate thresholds. When we analyze hybrid regimes combining Δ Centrality with the structural presumption, we hold the concentration threshold defined by the presumption fixed and optimize only over τ . Optimization is conducted separately for each enforcement regime and each decision rule. The analysis relies on demand estimates from the RCNL model.

Table 5 summarizes our results.³⁴ Across regimes, screening mergers based on changes in centrality performs well relative to standard screening tools. The gains are especially pronounced relative to GUPPI and the current concentration-based presumption, while the comparison with a fixed $\Delta HHI > 100$ threshold varies across enforcement regimes.

In the strict regime, where assumed cost efficiencies reduce marginal costs by 2%, centrality-based screening approves fewer mergers than either the CBP or GUPPI. If agencies relied exclusively on the CBP, they would approve approximately 1,350 more mergers than under Δ Centrality, whereas if they relied exclusively on GUPPI, they would approve approximately 2,809 more mergers. This stronger enforcement substantially reduces type II errors: relative to the CBP, the type II error rate falls by 24.9 percentage points, while relative to GUPPI it falls by 57.7 percentage points. These translate into significant welfare gains: since welfare-reducing mergers account

structural presumptions for identifying mergers with large price effects (Illanes et al., 2025).

³⁴We abstract from the dynamic effects of enforcement regimes. In practice, stricter regimes may affect consumer surplus by deterring merger activity, potentially discouraging both harmful and beneficial transactions. Moreover, we implicitly assume that merging parties will pass through realized efficiencies, though this may not necessarily be the case in practice (Harrington, 2021; Leccese et al., 2025).

Table 5: Merger Screening with Δ Centrality

	Proposed mergers	Allowed mergers (%)	Optimal CS (\$000,000)	Δ CS for allowed mergers (\$000,000)	Type I Errors (%)	Type I loss (\$000,000)	Type II Errors (%)	Type II loss (\$000,000)
Panel A: Strict Regime (2% assumed efficiency)								
Δ Centrality (%)	21,779	86.3	170948.323	151296.051	4.3	9697.895	37.5	9954.376
$\widehat{\Delta}$ Centrality (%)	21,779	84.8	170948.323	138348.206	6.1	16593.836	37.5	16006.281
Δ HHI > 100	21,779	86.9	170948.323	157053.038	2.9	6006.889	33.3	7888.396
GUPPI > 5%	21,779	99.2	170948.323	63831.848	0.0	0.000	95.2	107116.475
CBP	21,779	92.5	170948.323	142683.677	1.7	2986.926	62.4	25277.720
Panel B: Baseline Regime (5% assumed efficiency)								
Δ Centrality (%)	21,786	97.0	620883.345	606929.743	0.9	4189.995	52.0	9763.606
$\widehat{\Delta}$ Centrality (%)	21,786	96.4	620883.345	589141.499	1.8	13227.374	57.6	18514.472
Δ HHI > 100	21,786	86.8	620883.345	543210.085	9.3	77009.188	8.2	664.072
GUPPI > 5%	21,786	99.2	620883.345	590597.737	0.0	0.000	83.4	30285.609
CBP	21,786	92.5	620883.345	576552.279	4.6	41762.707	32.7	2568.360
Panel C: Lax Regime (10% assumed efficiency)								
Δ Centrality (%)	21,786	99.0	1538831.648	1528958.726	0.5	4996.928	38.5	4875.995
$\widehat{\Delta}$ Centrality (%)	21,786	99.2	1538831.648	1512876.638	0.5	6399.134	67.0	19555.877
Δ HHI > 100	21,786	86.8	1538831.648	1240871.979	12.4	297955.703	0.5	3.966
GUPPI > 5%	21,786	99.2	1538831.648	1536470.318	0.1	319.831	25.0	2041.499
CBP	21,786	92.5	1538831.648	1359206.126	6.7	179594.979	2.5	30.542

Notes: The table shows the number of proposed mergers (i.e., those that are profitable), the percentage of deals allowed, the maximum possible CS gains and the total change in CS for allowed mergers, the probability of blocking a welfare-enhancing deal (type I error) and that of approving a welfare-reducing deal (type II error), as well as the CS loss associated with each of them. We present three different regimes: strict, which assumes a 2% marginal cost efficiency (Panel A); baseline, which assumes a 5% marginal cost efficiency (Panel B); and lax, which assumes a 10% marginal cost efficiency (Panel C). We consider five screening approaches, and the thresholds for the Δ Centrality and $\widehat{\Delta}$ Centrality are optimally set for each regime-decision rule pair to maximize the total CS after the merger.

for 16.07% of proposed mergers, Δ Centrality implies an average consumer surplus loss of approximately \$7.575 million a year per mistakenly approved harmful merger, against the \$11.568 million a year and the \$32.128 million a year implied by CBP and GUPPI, respectively. As a benchmark, [Nevo \(2000\)](#) estimates that, in the ready-to-eat cereal industry, the Post-Nabisco merger, which was approved, reduced consumer surplus by approximately \$14 million a year.

These gains from Δ Centrality come at only moderate costs in terms of type I errors. As a result, Δ Centrality increases consumer surplus from allowed mergers by roughly 6% relative to the CBP and 137% relative to GUPPI. This corresponds to aggregate gains of \$8.6 billion and \$87.5 billion, respectively, or approximately \$0.46 million and \$4.65 million per merger approved under Δ Centrality.³⁵ In this strict regime,

³⁵All consumer surplus amounts in Table 5 are summed across the full set of simulated merger counterfactuals and should not be interpreted as the welfare effect of mergers occurring jointly.

the fixed $\Delta HHI > 100$ rule performs on average better than Δ Centrality, generating fewer type I and type II errors. However, the welfare gains are not particularly large: relative to the optimal rule that only approves welfare-enhancing deals, the ΔHHI rule preserves 91.9% of the total consumer surplus generated across all mergers against the 88.5% of the Δ Centrality rule. At the same time, the centrality rule still captures some harmful mergers that the concentration screen misses: for example, in France in 1985, the Fiat–Ford and Ford–Peugeot mergers both fall below the $\Delta HHI > 100$ threshold but are flagged by Δ Centrality and, assuming a 2% efficiency assumption, reduce consumer surplus by \$8.8 and \$8.1 million, respectively.³⁶

In the baseline regime, where assumed cost efficiencies reduce marginal costs by 5%, Δ Centrality delivers the best overall performance among the standalone screening rules reported in the table. Relative to the CBP, it approves approximately 980 additional mergers and reduces the type I error rate from 4.6% to 0.9%, lowering the associated aggregate consumer surplus loss from \$41.8 billion to \$4.2 billion. This comes at the cost of a higher type II error rate, which rises from 32.7% to 52.0%. However, the welfare gains from avoiding the mistaken blocking of beneficial mergers outweigh the losses from the additional harmful mergers that are approved. Overall, Δ Centrality raises consumer surplus from allowed mergers by \$30.4 billion relative to the CBP, a 5.3% increase, equivalent to approximately \$1.44 million per merger approved under Δ Centrality. Relative to GUPPI, it reduces type II errors by 31.4 percentage points and increases aggregate consumer surplus from allowed mergers by 2.8%. The comparison with $\Delta HHI > 100$ is even sharper: this rule blocks a much larger share of welfare-enhancing mergers, preserving 87.5% of the total consumer surplus generated by the optimal rule across all allowed mergers against the 97.8% of the Δ Centrality rule.

In the lax regime, where assumed cost efficiencies reduce marginal costs by 10%, the main screening concern shifts toward over-enforcement. The CBP and the $\Delta HHI > 100$ rule continue to block a sizable share of proposed mergers, even though most mergers are now welfare-enhancing. As a result, their aggregate type I losses become very large: \$179.6 billion under the CBP and \$298.0 billion under $\Delta HHI > 100$, compared with only \$5.0 billion under Δ Centrality. Centrality-based screening, therefore, increases consumer surplus from allowed mergers by 12.5% and 23.2% relative to the CBP and $\Delta HHI > 100$, respectively. Instead, the more permissive GUPPI performs slightly better than Δ Centrality in this regime.

From a practical standpoint, analyzing the performance of $\widehat{\Delta}$ Centrality, which only requires product characteristics, is particularly relevant. Its performance is naturally

³⁶Table C.3 provides details on these illustrative mergers, highlighting the role of diversion between the merging parties and, with respect to the rest of the network, in triggering the centrality-based screen.

weaker than that of the model-implied $\Delta\text{Centrality}$, but it remains highly informative. In the baseline regime, $\Delta\text{Centrality}$ increases consumer surplus from allowed mergers by 2.2% relative to the CBP and 8.5% relative to $\Delta HHI > 100$. In the lax regime, it delivers gains of 11.3% relative to the CBP and 21.9% relative to $\Delta HHI > 100$. Its performance is worse in the strict regime, where it is dominated by both the CBP and the $\Delta HHI > 100$ rule, suggesting that measurement error in centrality can be costly for mergers with relatively small expected efficiencies.

Taken together, these results suggest that centrality-based measures can usefully complement traditional merger screens. A fixed ΔHHI threshold performs well in the strict regime, but becomes overly restrictive as assumed efficiencies increase.³⁷ By contrast, centrality-based screening delivers more balanced performance across regimes, significantly improving on the CBP in the baseline and lax cases and outperforming GUPPI whenever approving harmful mergers is a central concern.³⁸

Finally, we note that there are also reasons to view the comparison with concentration-based rules in this setting as conservative for centrality-based screening. First, the automobile industry has relatively clear market segments, so product substitution patterns may already be reasonably well summarized by market shares. This limits the scope for $\Delta\text{Centrality}$ to improve on concentration-based measures. Because centrality incorporates product proximity in characteristic space, it may be more informative in industries where market boundaries are blurrier and substitution patterns are more heterogeneous. Second, the reference model used for the screening analysis in Table 5 is the RCNL. In logit-type models, substitution patterns are closely tied to market shares, which gives HHI-based screening measures a natural advantage. Centrality measures, by contrast, rely on distances in product characteristics. Under alternative demand systems in which substitution is less tightly linked to shares, centrality-based screens may offer larger improvements over traditional concentration thresholds.

6 Conclusion

This paper develops a tractable framework for analyzing horizontal mergers and acquisitions in differentiated-product industries using product networks. The key object is the diversion network: products are connected according to how demand is reallocated after a price change, and firms' pricing incentives depend on their position in the network. In this framework, equilibrium markups admit an additive decomposition

³⁷Table C.2 shows that an optimally chosen ΔHHI threshold performs comparably to $\Delta\text{Centrality}$ in this application, and in some cases slightly better. However, this requires using a threshold of 391 (896) in the baseline (lax) regime.

³⁸The hybrid rules in Table C.2 reinforce this interpretation: combining the CBP with $\Delta\text{Centrality}$ delivers performance close to the better of the two standalone rules in each regime.

into a monopolistic component and a Bonacich-centrality component. Mergers matter because they change how firms internalize diversion and therefore reshape the network that governs competitive pressure.

This perspective provides a unified characterization of the unilateral effects of acquisitions. We show that the price effects of a change in ownership can be decomposed into two forces: a direct upward-pricing-pressure term, familiar from first-order approaches, and a network-propagation term that captures how this initial pricing pressure spills over through the system of differentiated products. The first term depends on internalized diverted margins between the merging parties; the second depends on the structure of the broader diversion network. Hence, the framework extends first-order approaches by accounting for the indirect price responses that arise as pricing pressure propagates across competing products.

For full mergers, these direct and indirect effects aggregate into a change in centrality, which summarizes the unilateral effects of the acquisition and can therefore serve as a screening statistic for harmful mergers. The same structure also delivers closed-form expressions for consumer surplus changes and for the compensating marginal cost reductions needed to offset the anti-competitive effects of acquisitions.

Moreover, because under linear-quadratic demand diversion ratios are weighted inner products of product attributes, the diversion network can be directly mapped into product characteristics. This gives our approach practical appeal: unlike detailed price-quantity data, product characteristics are often readily observable and can increasingly be collected at scale, including from unstructured sources with the help of LLMs. Additionally, similar to traditional first-order approaches but unlike concentration-based screens, our framework does not require an ex-ante market definition, and it extends naturally to settings with partial ownership and common ownership where concentration-based measures are less straightforward to interpret.

In our application to the European automobile industry, the network model performs well despite not imposing any substitution nests ex ante. It generates substitution patterns and simulated merger impacts that are highly comparable to those from an RCNL model. Across simulated mergers, changes in centrality are strongly associated with predicted price increases and losses in consumer surplus. Regressing simulated price effects on alternative screening tools reveals that centrality explains a substantial share of the variation and remains informative relative to both concentration-based and traditional first-order measures. Finally, in screening exercises where agencies are assumed to maximize expected consumer surplus from approved mergers, we show that centrality-based rules tend to improve on the concentration-based presumption introduced by the 2023 Merger Guidelines and on a first-order approach based on GUPPI. However, the gains relative to rules based entirely on changes in HHI depend

on the assumed efficiencies: while centrality-based screening is preferred when expected efficiencies are larger, with small expected efficiencies of 2%, approving mergers with ΔHHI below 100 performs better.

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A Proofs

A.1 Proof of Lemma 1

Proof. This is the special case of Lemma 1' with $H^\varepsilon = I_J$, so $D_{H^\varepsilon} = D$ and the stated expression follows immediately. ■

A.2 Proof of Lemma 1'

Proof. Fix the local linearization and suppress the superscript ε for notational convenience. The locally linear demand system can be written as $\mathbf{q} = -S(\boldsymbol{\alpha} - \mathbf{p})$. Let $\mathbf{m} := \mathbf{p} - \mathbf{c}$.

Under the decision-weight matrix H , the decision maker for product j chooses p_j to maximize

$$\max_{p_j} \sum_{\ell=1}^J H_{j\ell} (p_\ell - c_\ell) q_\ell(\mathbf{p}),$$

taking p_{-j} as given. Differentiating the objective with respect to p_j gives

$$\begin{aligned} 0 &= \frac{\partial}{\partial p_j} \sum_{\ell=1}^J H_{j\ell} (p_\ell - c_\ell) q_\ell(\mathbf{p}) \\ &= \sum_{\ell=1}^J H_{j\ell} \left[\underbrace{\frac{\partial(p_\ell - c_\ell)}{\partial p_j}}_{=\mathbf{1}\{\ell=j\}} q_\ell(\mathbf{p}) + (p_\ell - c_\ell) \underbrace{\frac{\partial q_\ell(\mathbf{p})}{\partial p_j}}_{=: S_{\ell j}(\mathbf{p})} \right] \\ &= H_{jj} q_j(\mathbf{p}) + \sum_{\ell=1}^J H_{j\ell} (p_\ell - c_\ell) S_{\ell j}(\mathbf{p}). \end{aligned}$$

Using $H_{jj} = 1$, the j -th first-order condition can be written as

$$q_j(\mathbf{p}) + \sum_{\ell=1}^J H_{j\ell} S_{\ell j}(\mathbf{p}) m_\ell = 0, \quad j = 1, \dots, J. \quad (41)$$

By symmetry of S , the stacked FOCs are

$$\mathbf{q}(\mathbf{p}) + (H \odot S(\mathbf{p}))(\mathbf{p} - \mathbf{c}) = 0. \quad (42)$$

Substituting $\mathbf{q} = -S(\boldsymbol{\alpha} - \mathbf{p})$ into (42) and writing $\mathbf{p} = \mathbf{c} + \mathbf{m}$ gives

$$-S(\boldsymbol{\alpha} - \mathbf{c} - \mathbf{m}) + (H \odot S)\mathbf{m} = 0 \iff -S(\boldsymbol{\alpha} - \mathbf{c}) + (S + H \odot S)\mathbf{m} = 0.$$

Premultiplying by $(H \odot S)^{-1}$ and using

$$D_H = I_J - (H \odot S)^{-1}S \implies (H \odot S)^{-1}S = I_J - D_H,$$

we obtain

$$(2I_J - D_H)\mathbf{m} = (I_J - D_H)(\boldsymbol{\alpha} - \mathbf{c}).$$

Hence

$$\mathbf{m} = (2I_J - D_H)^{-1} (I_J - D_H)(\boldsymbol{\alpha} - \mathbf{c}) = \left(I_J - \frac{1}{2}D_H\right)^{-1} (I_J - D_H)\frac{\boldsymbol{\alpha} - \mathbf{c}}{2}.$$

Let $\mathbf{u} := (\boldsymbol{\alpha} - \mathbf{c})/2$. Since

$$\mathbf{b}\left(D_H, \frac{1}{2}, \mathbf{u}\right) = \left(I_J - \frac{1}{2}D_H\right)^{-1} \frac{1}{2}D_H \mathbf{u},$$

we have

$$\mathbf{u} - \mathbf{b}\left(D_H, \frac{1}{2}, \mathbf{u}\right) = \left(I_J - \frac{1}{2}D_H\right)^{-1} (I_J - D_H)\mathbf{u} = \mathbf{m}.$$

Therefore

$$\mathbf{p} - \mathbf{c} = \frac{\boldsymbol{\alpha} - \mathbf{c}}{2} - \mathbf{b}\left(D_H, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right),$$

which is the claimed expression (and replacing $(\boldsymbol{\alpha}, \mathbf{c}, H)$ by their ε -perturbed counterparts yields the stated ε version). ■

A.3 Proof of Proposition 1

Proof. This is Proposition 1' specialized to $H = I_J$ and $\Delta H = E_{12}$, for which $D_H = D$ and $\boldsymbol{\omega}_{\Delta H}(H) = D_{12}(p_2 - c_2)\mathbf{e}_1$, and Roy's identity then gives the CS expression. ■

A.4 Proof of Proposition 1'

Proof. Consider $H^\varepsilon = H + \varepsilon\Delta H$, with $\boldsymbol{\alpha}, \mathbf{c}, S = \frac{d\mathbf{q}}{d\mathbf{p}'}$ fixed under local linearization. The locally linearized demand is

$$\mathbf{q}(\mathbf{p}^\varepsilon) = -S(\boldsymbol{\alpha} - \mathbf{p}^\varepsilon).$$

The FOC is

$$\mathbf{q}(\mathbf{p}^\varepsilon) + (H^\varepsilon \odot S)(\mathbf{p}^\varepsilon - \mathbf{c}) = 0,$$

or equivalently

$$-S(\boldsymbol{\alpha} - \mathbf{p}^\varepsilon) + (H^\varepsilon \odot S)(\mathbf{p}^\varepsilon - \mathbf{c}) = 0.$$

Differentiating at $\varepsilon = 0$ gives

$$S\dot{\mathbf{p}} + (\Delta H \odot S)(\mathbf{p} - \mathbf{c}) + (H \odot S)\dot{\mathbf{p}} = 0,$$

so

$$(S + H \odot S)\dot{\mathbf{p}} = -(\Delta H \odot S)(\mathbf{p} - \mathbf{c}).$$

Premultiplying by $(H \odot S)^{-1}$ and using $(H \odot S)^{-1}S = I_J - D_H$ yields

$$(2I_J - D_H)\dot{\mathbf{p}} = -(H \odot S)^{-1}[(\Delta H \odot S)(\mathbf{p} - \mathbf{c})] = \boldsymbol{\omega}_{\Delta H}(H).$$

Therefore

$$\dot{\mathbf{p}} = (2I_J - D_H)^{-1} \boldsymbol{\omega}_{\Delta H}(H) = \frac{1}{2} \left(I_J - \frac{1}{2} D_H \right)^{-1} \boldsymbol{\omega}_{\Delta H}(H) = \frac{1}{2} \boldsymbol{\kappa} \left(D_H, \frac{1}{2}, \boldsymbol{\omega}_{\Delta H}(H) \right),$$

which is (12). By Roy's identity, evaluated at the baseline equilibrium,

$$\left. \frac{\partial CS}{\partial \varepsilon} \right|_{\varepsilon=0} = -\mathbf{q}' \dot{\mathbf{p}} = -\frac{1}{2} \mathbf{q}' \boldsymbol{\kappa} \left(D_H, \frac{1}{2}, \boldsymbol{\omega}_{\Delta H}(H) \right),$$

which is (13). ■

A.5 Proof of Proposition 2

Proof. This is Proposition 2' specialized to $H = I_J$, $\Delta H = E_{12}$, and $\left. \frac{dc_1}{d\varepsilon} \right|_0 = \left. \frac{dc_2}{d\varepsilon} \right|_0 = \frac{dc}{d\varepsilon}$ with all other cost reductions equal to zero, together with $\boldsymbol{\omega}_{E_{12}}(I_J) = D_{12}(p_2 - c_2)\mathbf{e}_1$ from Proposition 1. ■

A.6 Proof of Proposition 2'

Proof. Let $H^\varepsilon = H + \varepsilon \Delta H$ and write post-change costs as $\mathbf{c}^\varepsilon = \mathbf{c} - \varepsilon \mathbf{s}$, where s_j is a marginal cost reduction. The locally linearized demand is $\mathbf{q}(\mathbf{p}^\varepsilon) = -S(\boldsymbol{\alpha} - \mathbf{p}^\varepsilon)$, so the FOC is

$$-S(\boldsymbol{\alpha} - \mathbf{p}^\varepsilon) + (H^\varepsilon \odot S)(\mathbf{p}^\varepsilon - \mathbf{c}^\varepsilon) = 0.$$

Differentiating at $\varepsilon = 0$:

$$S\dot{\mathbf{p}} + (\Delta H \odot S)(\mathbf{p} - \mathbf{c}) + (H \odot S)(\dot{\mathbf{p}} + \mathbf{s}) = 0.$$

Hence

$$(S + H \odot S)\dot{\mathbf{p}} = -(\Delta H \odot S)(\mathbf{p} - \mathbf{c}) - (H \odot S)\mathbf{s}.$$

Premultiplying by $(H \odot S)^{-1}$:

$$(2I_J - D_H)\dot{\mathbf{p}} = \boldsymbol{\omega}_{\Delta H}(H) - \mathbf{s},$$

so

$$\dot{\mathbf{p}} = \frac{1}{2} \boldsymbol{\kappa}\left(D_H, \frac{1}{2}, \boldsymbol{\omega}_{\Delta H}(H) - \mathbf{s}\right).$$

CS-neutrality requires $0 = \mathbf{q}'\dot{\mathbf{p}}$, therefore

$$\mathbf{q}'\boldsymbol{\kappa}\left(D_H, \frac{1}{2}, \boldsymbol{\omega}_{\Delta H}(H)\right) = \mathbf{q}'\boldsymbol{\kappa}\left(D_H, \frac{1}{2}, \mathbf{s}\right).$$

By linearity in node weights and $\mathbf{s} = \sum_{j=1}^J s_j \mathbf{e}_j$,

$$\mathbf{q}'\boldsymbol{\kappa}\left(D_H, \frac{1}{2}, \mathbf{s}\right) = \sum_{j=1}^J s_j \mathbf{q}'\boldsymbol{\kappa}(j; D_H),$$

which gives the stated condition. ■

A.7 Proof of Proposition 3

Proof. This is Proposition 3' specialized to $H = I_J$, $\widehat{H} = M_{12}$, and (for (10)) a common cost reduction vector $\Delta \mathbf{c} = \Delta c (\mathbf{e}_1 + \mathbf{e}_2)$. ■

A.8 Proof of Proposition 3'

Proof. From Lemma 1', with no cost synergy,

$$\mathbf{p}(H) - \mathbf{c} = \frac{\boldsymbol{\alpha} - \mathbf{c}}{2} - \mathbf{b}\left(D_H, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right),$$

$$\mathbf{p}(\widehat{H}) - \mathbf{c} = \frac{\boldsymbol{\alpha} - \mathbf{c}}{2} - \mathbf{b}\left(D_{\widehat{H}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right).$$

Subtracting gives

$$\Delta \mathbf{p} := \mathbf{p}(\widehat{H}) - \mathbf{p}(H) = \mathbf{b}\left(D_H, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right) - \mathbf{b}\left(D_{\widehat{H}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right),$$

which is (15).

Now add a cost-reduction vector $\Delta \mathbf{c}$ so post-change costs are $\mathbf{c} - \Delta \mathbf{c}$ under $\widehat{H}$. Again by Lemma 1',

$$\begin{aligned} \mathbf{p}(\widehat{H}, \Delta \mathbf{c}) - (\mathbf{c} - \Delta \mathbf{c}) &= \frac{\boldsymbol{\alpha} - (\mathbf{c} - \Delta \mathbf{c})}{2} - \mathbf{b}\left(D_{\widehat{H}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - (\mathbf{c} - \Delta \mathbf{c})}{2}\right) \\ &= \frac{\boldsymbol{\alpha} - \mathbf{c} + \Delta \mathbf{c}}{2} - \mathbf{b}\left(D_{\widehat{H}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c} + \Delta \mathbf{c}}{2}\right). \end{aligned}$$

Using linearity of $\mathbf{b}$ in node weights and $\mathbf{p}(\widehat{H})$ above,

$$\mathbf{p}(\widehat{H}, \Delta \mathbf{c}) - \mathbf{p}(\widehat{H}) = -\frac{\Delta \mathbf{c}}{2} - \mathbf{b}\left(D_{\widehat{H}}, \frac{1}{2}, \frac{\Delta \mathbf{c}}{2}\right) = -\frac{1}{2} \boldsymbol{\kappa}\left(D_{\widehat{H}}, \frac{1}{2}, \Delta \mathbf{c}\right).$$

Therefore total price change relative to the baseline H is

$$\Delta \mathbf{p}_{\text{total}} = \mathbf{b}\left(D_H, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right) - \mathbf{b}\left(D_{\widehat{H}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right) - \frac{1}{2} \boldsymbol{\kappa}\left(D_{\widehat{H}}, \frac{1}{2}, \Delta \mathbf{c}\right).$$

Imposing CS-neutrality in the quantity-weighted approximation, $\mathbf{q}' \Delta \mathbf{p}_{\text{total}} = 0$, gives

$$\frac{1}{2} \mathbf{q}' \boldsymbol{\kappa}\left(D_{\widehat{H}}, \frac{1}{2}, \Delta \mathbf{c}\right) = \mathbf{q}' \left[\mathbf{b}\left(D_H, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right) - \mathbf{b}\left(D_{\widehat{H}}, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right) \right].$$

Finally, by linearity and $\Delta \mathbf{c} = \sum_{j=1}^J \Delta c_j \mathbf{e}_j$,

$$\mathbf{q}' \boldsymbol{\kappa}\left(D_{\widehat{H}}, \frac{1}{2}, \Delta \mathbf{c}\right) = \sum_{j=1}^J \mathbf{q}' \boldsymbol{\kappa}(j; D_{\widehat{H}}) \Delta c_j,$$

which is (16). ■

A.9 Proof of Proposition 4

Proof. From (18),

$$\mathbf{q}(\mathbf{p}) = \frac{1}{\beta} (I_J + \eta X X')^{-1} (\boldsymbol{\alpha} - \mathbf{p}).$$

By the Woodbury matrix identity:

$$(I_J + \eta X X')^{-1} = I_J - X \left(\frac{1}{\eta} I_K + X' X \right)^{-1} X' = I_J - \Theta,$$

where $\Theta = X \Omega^{-1} X'$ and $\Omega = \frac{1}{\eta} I_K + X' X$. Therefore

$$\mathbf{q}(\mathbf{p}) = \frac{1}{\beta} (I_J - \Theta) (\boldsymbol{\alpha} - \mathbf{p}),$$

which is (20). Taking the j -th row yields

$$q_j(\mathbf{p}) = a_j - \frac{1}{\beta} (1 - \theta_{jj}) p_j + \frac{1}{\beta} \sum_{\ell \neq j} \theta_{j\ell} p_\ell,$$

with $\mathbf{a} = \frac{1}{\beta} (I_J - \Theta) \boldsymbol{\alpha}$. ■

A.10 Proof of Proposition 5

Proof. By definition,

$$\theta_{j\ell} = \mathbf{x}'_j \Omega^{-1} \mathbf{x}_\ell, \quad \Omega = \frac{1}{\eta} I_K + X'X.$$

Let $X'X = U\Lambda^{X'X}U'$, where $U = [u_1, \dots, u_K]$ is orthogonal and $\Lambda^{X'X} = \text{diag}(\lambda_1^{X'X}, \dots, \lambda_K^{X'X})$.

Then

$$\Omega^{-1} = U \left(\frac{1}{\eta} I_K + \Lambda^{X'X} \right)^{-1} U'.$$

Therefore

$$\theta_{j\ell} = \mathbf{x}'_j U \left(\frac{1}{\eta} I_K + \Lambda^{X'X} \right)^{-1} U' \mathbf{x}_\ell.$$

Define $\tilde{\mathbf{x}}_j := U' \mathbf{x}_j$, so $\tilde{x}_{jk} = u'_k \mathbf{x}_j$. Since $\left(\frac{1}{\eta} I_K + \Lambda^{X'X} \right)^{-1}$ is diagonal, we get

$$\theta_{j\ell} = \sum_{k=1}^K \frac{\tilde{x}_{jk} \tilde{x}_{\ell k}}{1/\eta + \lambda_k^{X'X}},$$

which is (21). ■

B Equilibrium Existence and Uniqueness

In this appendix, we provide a sufficient condition for existence and uniqueness of an interior equilibrium for the linear-quadratic utility model under a general control matrix H .

Proposition 6 *Suppose $\beta > 0$, $\alpha_j > c_j$ for all j , and fix a control matrix H with $H_{jj} = 1$ and $0 \leq H_{jk} \leq 1$ for all $j \neq k$. If*

$$(1 - \theta_{jj})(\alpha_j - c_j) > \sum_{k \neq j} |\theta_{jk}| (\alpha_k - c_k) \quad \text{for all } j, \quad (43)$$

then there exists a unique Nash equilibrium $\mathbf{p}^$, and it is interior—i.e., $p_j^* \in (c_j, \alpha_j)$ for all j .*

Proof. Let

$$K := \prod_{j=1}^J [c_j, \alpha_j].$$

Since $\alpha_j > c_j$ for all j , the set K is nonempty and closed. Endow $\mathbb{R}^J$ with the weighted sup norm

$$\|\mathbf{x}\|_* := \max_j \frac{|x_j|}{\alpha_j - c_j}.$$

Because K is a closed subset of the finite-dimensional normed space $(\mathbb{R}^J, \|\cdot\|_*)$, it is complete.

Under the control matrix H , product j chooses p_j to maximize

$$\sum_{\ell=1}^J H_{j\ell}(p_\ell - c_\ell)q_\ell(\mathbf{p}).$$

The first-order condition with respect to p_j is

$$0 = q_j(\mathbf{p}) + \sum_{\ell=1}^J H_{j\ell}(p_\ell - c_\ell) \frac{\partial q_\ell}{\partial p_j}.$$

Because $I_J - \Theta$ is symmetric, we have $\theta_{j\ell} = \theta_{\ell j}$ and hence

$$\frac{\partial q_j}{\partial p_j} = -\frac{1}{\beta}(1 - \theta_{jj}), \quad \frac{\partial q_\ell}{\partial p_j} = \frac{1}{\beta}\theta_{j\ell} \quad (\ell \neq j).$$

Substituting these expressions into the first-order condition and multiplying through by $\beta > 0$, we obtain

$$0 = (1 - \theta_{jj})(\alpha_j - p_j) - \sum_{k \neq j} \theta_{jk}(\alpha_k - p_k) - (1 - \theta_{jj})(p_j - c_j) + \sum_{k \neq j} H_{jk}\theta_{jk}(p_k - c_k).$$

Rearranging gives the best-reply map

$$T_j(\mathbf{p}) := \frac{\alpha_j + c_j}{2} - \frac{1}{2(1 - \theta_{jj})} \sum_{k \neq j} \theta_{jk} \left[(\alpha_k - c_k) - (1 + H_{jk})(p_k - c_k) \right]. \quad (44)$$

Moreover, product j 's objective is strictly concave in its own price, since

$$\frac{\partial^2}{\partial p_j^2} \sum_{\ell=1}^J H_{j\ell}(p_\ell - c_\ell) q_\ell(\mathbf{p}) = -\frac{2}{\beta}(1 - \theta_{jj}) < 0.$$

Thus $T_j(\mathbf{p})$ is the unique best reply of product j to $\mathbf{p}_{-j}$.

We first show that $T(K) \subseteq K$. Take any $\mathbf{p} \in K$. Then for every k ,

$$0 \leq p_k - c_k \leq \alpha_k - c_k.$$

Since $0 \leq H_{jk} \leq 1$, it follows that

$$-H_{jk}(\alpha_k - c_k) \leq (\alpha_k - c_k) - (1 + H_{jk})(p_k - c_k) \leq \alpha_k - c_k,$$

and therefore

$$|(\alpha_k - c_k) - (1 + H_{jk})(p_k - c_k)| \leq \alpha_k - c_k.$$

Using (44), we obtain

$$\begin{aligned} \left| T_j(\mathbf{p}) - \frac{\alpha_j + c_j}{2} \right| &\leq \frac{1}{2(1 - \theta_{jj})} \sum_{k \neq j} |\theta_{jk}| |(\alpha_k - c_k) - (1 + H_{jk})(p_k - c_k)| \\ &\leq \frac{1}{2(1 - \theta_{jj})} \sum_{k \neq j} |\theta_{jk}| (\alpha_k - c_k) \\ &< \frac{\alpha_j - c_j}{2}, \end{aligned}$$

where the last inequality follows from (43). Hence

$$c_j < T_j(\mathbf{p}) < \alpha_j \quad \text{for all } j.$$

Therefore

$$T(K) \subseteq \prod_{j=1}^J (c_j, \alpha_j) \subseteq K.$$

We next show that T is a contraction on K . For any $\mathbf{p}, \tilde{\mathbf{p}} \in K$, (44) implies

$$T_j(\mathbf{p}) - T_j(\tilde{\mathbf{p}}) = \frac{1}{2(1 - \theta_{jj})} \sum_{k \neq j} (1 + H_{jk}) \theta_{jk} (p_k - \tilde{p}_k).$$

Hence

$$\begin{aligned} \frac{|T_j(\mathbf{p}) - T_j(\tilde{\mathbf{p}})|}{\alpha_j - c_j} &\leq \frac{1}{2(1 - \theta_{jj})(\alpha_j - c_j)} \sum_{k \neq j} (1 + H_{jk}) |\theta_{jk}| |p_k - \tilde{p}_k| \\ &\leq \frac{1}{2(1 - \theta_{jj})(\alpha_j - c_j)} \sum_{k \neq j} (1 + H_{jk}) |\theta_{jk}| (\alpha_k - c_k) \|\mathbf{p} - \tilde{\mathbf{p}}\|_*. \end{aligned}$$

Since $0 \leq H_{jk} \leq 1$, we have $(1 + H_{jk})/2 \leq 1$, so

$$\begin{aligned} \frac{|T_j(\mathbf{p}) - T_j(\tilde{\mathbf{p}})|}{\alpha_j - c_j} &\leq \frac{\sum_{k \neq j} |\theta_{jk}| (\alpha_k - c_k)}{(1 - \theta_{jj})(\alpha_j - c_j)} \|\mathbf{p} - \tilde{\mathbf{p}}\|_* \\ &< \|\mathbf{p} - \tilde{\mathbf{p}}\|_*, \end{aligned}$$

where the strict inequality again follows from (43). Therefore there exists $\rho \in (0, 1)$ such that

$$\|T(\mathbf{p}) - T(\tilde{\mathbf{p}})\|_* \leq \rho \|\mathbf{p} - \tilde{\mathbf{p}}\|_* \quad \text{for all } \mathbf{p}, \tilde{\mathbf{p}} \in K.$$

Thus T is a contraction on the complete metric space $(K, \|\cdot\|_*)$.

By Banach's fixed point theorem, T has a unique fixed point $\mathbf{p}^* \in K$. Since each T_j is the unique best reply of product j , this fixed point is the unique Nash equilibrium. Moreover, because $T(K) \subseteq \prod_{j=1}^J (c_j, \alpha_j)$, the fixed point satisfies $p_j^* \in (c_j, \alpha_j)$ for all j . ■

C Additional Details on Empirical Analyses

Segment	Own-price	Cross-price elasticity		Diversion ratio	
		Within segment	Across segments	Within segment	Across segments
Logit					
Subcompact	-1.32	0.00	0.00	0.05	0.05
Compact	-1.84	0.00	0.00	0.05	0.05
Intermediate	-2.17	0.00	0.00	0.04	0.04
Standard	-2.84	0.00	0.00	0.02	0.02
Luxury	-3.18	0.00	0.00	0.04	0.04
Nested Logit					
Subcompact	-2.25	0.03	0.00	1.30	0.05
Compact	-3.05	0.05	0.00	1.54	0.05
Intermediate	-2.77	0.00	0.00	0.04	0.03
Standard	-3.75	0.01	0.00	0.16	0.02
Luxury	-5.05	0.22	0.00	4.23	0.05
RC Logit					
Subcompact	-1.54	0.02	0.02	1.12	0.99
Compact	-2.15	0.02	0.02	1.06	1.05
Intermediate	-2.53	0.02	0.02	0.87	0.80
Standard	-3.32	0.02	0.02	0.72	0.53
Luxury	-3.69	0.05	0.02	1.33	0.96
RC Nested Logit					
Subcompact	-2.87	0.07	0.02	2.33	0.84
Compact	-3.43	0.08	0.02	2.24	0.85
Intermediate	-3.14	0.03	0.02	0.83	0.58
Standard	-4.30	0.03	0.02	0.85	0.38
Luxury	-5.34	0.29	0.02	4.97	1.57
Network					
Subcompact	-7.38	0.17	0.10	2.31	0.73
Compact	-9.36	0.22	0.09	2.35	0.83
Intermediate	-12.40	0.30	0.11	2.38	0.86
Standard	-20.40	0.51	0.13	2.51	0.89
Luxury	-19.85	0.62	0.11	3.19	1.00

Table C.1: Average elasticities and diversion ratios across all markets by segment. Own elasticity values are trimmed at the 10th percentile before averaging.

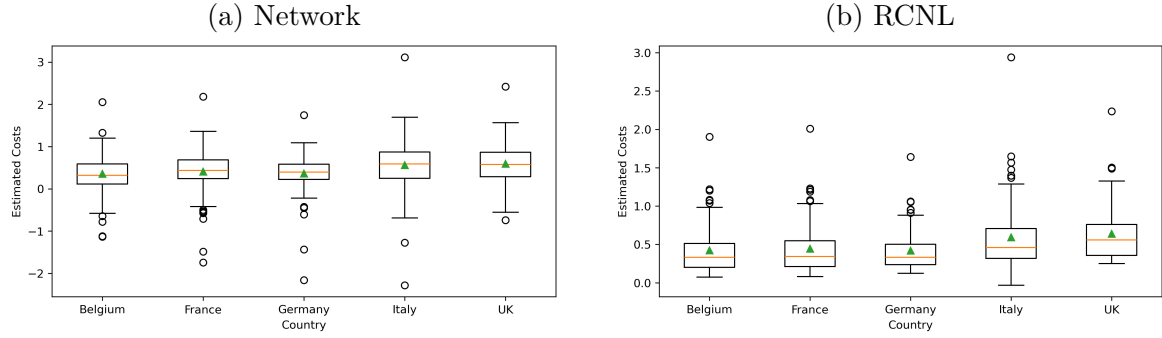


Figure C.1: Estimated marginal costs by country for 1999

Notes: The figure illustrates the distribution of the estimated marginal costs across countries for the network (left) and the RCNL (models) in 1999 using a box-plot.

Table C.2: Merger Screening with Δ Centrality

	Proposed mergers	Allowed mergers (%)	Optimal CS (\$000,000)	Δ CS for allowed mergers (\$000,000)	Type I Errors (%)	Type I loss (\$000,000)	Type II Errors (%)	Type II loss (\$000,000)
Panel B: Strict Regime (2% assumed efficiency)								
ΔHHI^*	21,779	87.4	170948.323	157388.785	2.6	5224.445	34.9	8335.092
CBP or % Δ Centrality	21,779	86.2	170948.323	151796.503	4.4	9991.997	36.9	9159.823
CBP & % Δ Centrality	21,779	92.7	170948.323	142183.225	1.6	2692.824	63.0	26072.274
Panel C: Baseline Regime (5% assumed efficiency)								
ΔHHI^*	21,786	97.1	620883.345	612206.955	0.5	2090.066	46.2	6586.324
CBP or % Δ Centrality	21,786	92.4	620883.345	576466.716	4.6	41865.199	31.6	2551.430
CBP & % Δ Centrality	21,786	97.1	620883.345	607015.306	0.8	4087.503	53.1	9780.536
Panel D: Lax Regime (10% assumed efficiency)								
ΔHHI^*	21,786	99.4	1538831.648	1533015.123	0.1	1075.939	44.5	4740.586
CBP or % Δ Centrality	21,786	92.5	1538831.648	1359206.126	6.7	179594.979	2.5	30.542
CBP & % Δ Centrality	21,786	99.0	1538831.648	1528958.726	0.5	4996.928	38.5	4875.995

Notes: The table shows the number of proposed mergers (i.e., those that are profitable), the percentage of deals allowed, the maximum possible CS gains and the total change in CS for allowed mergers, the probability of blocking a welfare-enhancing deal (type I error) and that of approving a welfare-reducing deal (type II error), as well as the CS loss associated with each of them. We present three different regimes: strict, which assumes a 2% marginal cost efficiency (Panel A); baseline, which assumes a 5% marginal cost efficiency (Panel B); and lax, which assumes a 10% marginal cost efficiency (Panel C). We consider three additional screening approaches using: two hybrid systems that combine changes in centrality with CBP and the ΔHHI rule implemented with optimal thresholds. The thresholds for ΔHHI^* and Δ Centrality are optimally set for each regime-decision rule pair to maximize the total CS after the merger.

C.1 Construction of Aggregate Centrality

To compute the Bonacich centrality of a merger, we first proceed by computing the vector of centralities for all products, as presented in Equation 1:

$$\mathbf{b}\left(D_{H^\varepsilon}, \frac{1}{2}, \frac{\boldsymbol{\alpha}^\varepsilon - \mathbf{c}^\varepsilon}{2}\right) = \left(I_J - \frac{1}{2}D_{H^\varepsilon}\right)^{-1} \frac{1}{2}D_{H^\varepsilon} \left(\frac{\boldsymbol{\alpha}^\varepsilon - \mathbf{c}^\varepsilon}{2}\right).$$

We then aggregate the individual centralities to our market level measure using the weighted sum of all of them, $\mathbf{q}'\mathbf{b}\left(D_H, \frac{1}{2}, \frac{\boldsymbol{\alpha} - \mathbf{c}}{2}\right)$.

To simulate a merger, we compute new centralities taking into account the new ownership structure, which then allows us to update prices, using Lemma 1. Then, we update quantities and market shares.

For a practical example, we consider the market in France, 1985, with two different proposed mergers, one between Fiat and Ford, and the other between Ford and Peugeot. Table C.3 presents some statistics regarding the mergers.

Table C.3: Characteristics and Implications of Selected Mergers

	Merger 1		Merger 2	
	Fiat	Ford	Ford	Peugeot
Merger characteristics				
Pre-merger HHI	1582.5		1582.5	
Post-merger HHI	1675.1		1671.7	
Δ HHI	92.1		88.4	
Δ Centrality	-0.347%		-0.358%	
$\Delta P\%$	0.465%		0.438%	
$\Delta P\%$ - Merged Firms	2.176%		2.037%	
Δ CS - 2% Efficiency (\$000,000)	-8.776		-8.079	
Δ CS - 5% Efficiency (\$000,000)	18.045		19.493	
Δ CS - 10% Efficiency (\$000,000)	64.620		67.379	
Number of Products	10	5	5	10
Quantity Sold	96,781	203,324	203,324	92,871
Market Share	4.68%	9.84%	9.84%	4.49%
Diversions statistics				
Average diversion to all products	0.59%	1.11%	1.11%	0.54%
Average diversion from all products	1.01%	0.90%	0.90%	0.98%
Average Cross-diversion to partner	1.01%	0.27%	0.27%	1.05%

The mergers presented are also examples of mergers that are not blocked by the CBP (Δ HHI < 100 and Post-merger HHI < 1800), but lead to a loss of consumer surplus (in the 2% assumed efficiency case). Both mergers are, however, blocked by our $\Delta\%$ Centrality measure ($\Delta\%$ Centrality < -0.322%). The use of the Bonacich

centrality allows us to incorporate information on diversion between products in a way that HHI cannot.

In particular, the diversion statistics in the bottom panel help illustrate the determinants of the centrality change. In our model, centrality is a competitive-pressure term: higher centrality lowers equilibrium prices, while a merger raises prices by internalizing cross-firm diversion links and thereby reducing centrality. The last row, average cross-diversion to the merger partner, captures the size of the links that are directly removed from the competitive-diversion matrix after the merger, and is therefore the object most closely related to GUPPI. The magnitude of the resulting decline in centrality, however, also depends on where those links sit in the broader diversion network. A lost diversion link has a larger effect when the originating firm is an important destination for diversion from other products, and when the merger partner is itself strongly connected to the rest of the market through its outward diversion patterns. The rows reporting average diversion from and to all products provide descriptive summaries of these two network features. Thus, cross-diversion determines the direct bilateral force, while the broader diversion environment determines how much removing those bilateral links reduces aggregate centrality.

C.2 Construction of Aggregate GUPPI

Because the Gross Upward Pricing Pressure Index (GUPPI) is typically defined at the product-level and in our analyses we consider mergers of multi-product firms, this section describes how we construct an aggregate version of the GUPPI. We start from the GUPPI for product j with respect to product ℓ , which is given by:

$$\text{GUPPI}_{j\ell} = \frac{p_\ell^{(0)} - c_\ell}{p_j^{(0)}} D_{j\ell}(\mathbf{p}^{(0)}), \quad (45)$$

where $p_r^{(0)}$ denotes the pre-merger price of product r , c_r denotes its marginal cost, and $D_{j\ell}(\mathbf{p}^{(0)})$ is the diversion ratio from product j to product ℓ , evaluated at pre-merger prices.

Since we consider mergers between multi-product firms, we aggregate product-level GUPPIs to obtain firm-level and merger-level measures. Consider two merging firms indexed by $f \in \{g, h\}$, with product sets

$$\mathcal{J}_f = \{1, \dots, J_f\}.$$

For a product $j \in \mathcal{J}_g$, its GUPPI with respect to all products of firm h is defined as

the sum of the individual GUPPIs:

$$\text{GUPPI}_{j \rightarrow \mathcal{J}_h} = \sum_{\ell \in \mathcal{J}_h} \frac{p_\ell^{(0)} - c_\ell}{p_j^{(0)}} D_{j\ell}(\mathbf{p}^{(0)}). \quad (46)$$

This measures the upward pricing pressure on product j generated by the products owned by the merger partner.

We then aggregate across all products owned by firm g using a sales-weighted average:

$$\text{GUPPI}_g = \sum_{j \in \mathcal{J}_g} \omega_j^g \text{GUPPI}_{j \rightarrow \mathcal{J}_h}, \quad (47)$$

where

$$\omega_j^g = \frac{\text{sales}_j}{\sum_{r \in \mathcal{J}_g} \text{sales}_r} \quad (48)$$

is product j 's share of firm g 's total sales.

Analogously, for firm h , we compute

$$\text{GUPPI}_h = \sum_{k \in \mathcal{J}_h} \omega_k^h \text{GUPPI}_{k \rightarrow \mathcal{J}_g}, \quad (49)$$

where ω_k^h is product k 's share of firm h 's total sales.

Finally, we aggregate the two firm-level GUPPIs into a merger-level GUPPI using firm-level sales weights:

$$\text{GUPPI}_{\text{merger}} = \sum_{f \in \{g, h\}} \Omega_f \text{GUPPI}_f, \quad (50)$$

where

$$\Omega_f = \frac{\sum_{j \in \mathcal{J}_f} \text{sales}_j}{\sum_{r \in \mathcal{J}_g} \text{sales}_r + \sum_{r \in \mathcal{J}_h} \text{sales}_r} \quad (51)$$

is firm f 's share of the merging parties' total sales. We use this final aggregation, $\text{GUPPI}_{\text{merger}}$, as our screening tool.